

Browning M1919A4 Machine Gun

PAGES FROM THE ORIGINAL MANUAL

FM 23-45, Basic Field Manual – Browning Machine Gun, Caliber .30, HB, M1919A4, Ground (War Department, 1943)

WHAT'S INSIDE

Chapter 1, Mechanical Training: description, taking it apart, cleaning, how it works, stoppages, the tripod, accessories, sights and ammunition.

65 pages from the original follow: the manual's cover, then the section about this weapon. Original US Government publication (public domain). Scans courtesy of the Internet Archive and Wikimedia Commons.

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WAR DEPARTMENT

BASIC FIELD MANUAL

**BROWNING MACHINE GUN,
CALIBER .30, HB, M1919A4,
GROUND**

April 12, 1943

BASIC FIELD MANUAL

BROWNING MACHINE GUN, CALIBER .30, HB M1919A4, GROUND

(This manual supersedes FM 23-45, August 14, 1940; C 1, August 19, 1941; and C 2, July 13, 1942.)

Caliber .30 Browning Machine Guns, HB, M1919A4, Ground, are being replaced by Browning Machine Guns, Caliber .30, HB, M1919A6, which are the M1919A4 guns with bipods, shoulder stocks, and carrying handles. Instructions and the necessary materials for modifying the weapons now in service will be issued. FM 23-45 will be revised accordingly.

Attention is directed to FM 21-7, for details as to how appropriate training films and film strips are intended to be used and how they are made available for use during training with the Browning Machine Gun, Caliber .30, M1919A4.

CHAPTER I

MECHANICAL TRAINING

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SECTION I

DESCRIPTION

■ 1. PRINCIPLE OF OPERATION.—The machine gun, caliber .30, M1919A4, is recoil-operated, belt-fed, and air-cooled. In recoil operation, the rearward force of the expanding powder gas (kick) furnishes the operating energy. The purpose of the front barrel bearing plug (see note on p. 2) is to insure the positive recoil of the barrel and barrel extension. The principle of operation is that gases expand equally in all directions. The front barrel bearing plug traps a certain

amount of gas in front of the muzzle. The force of the expanding gas acts on the muzzle, helping to force the heavy barrel to the rear. The moving parts, while locked together at the moment of the explosion, are left free within the receiver to be forced to the rear by the recoil. This movement is controlled by means of various springs, cams, and levers, and is utilized to perform the necessary mechanical operations of unlocking the breech, extracting and ejecting the empty case, feeding in a new round and loading, as well as cocking, locking, and firing the mechanism. The receiver mechanism is for all practical purposes the receiver of the Browning machine gun, M1917. (See fig. 1.)

NOTE.—FRONT BARREL BEARING PLUGS. There are three types of front barrel bearing plugs in use.

a. The M2-06 plug was made for the old gun for firing M2 or 06 ammunition. Identified by marking "M2-06."

b. The caliber .30, M1 plug with a slightly larger escape hole than the M2-06 plug, was made for the old gun for firing M1 ammunition. Identified by marking "cal. .30 M1."

c. The .718 plug was designed for the new (1941) gun with the stronger driving spring. Identified by marking "DIA. .718."

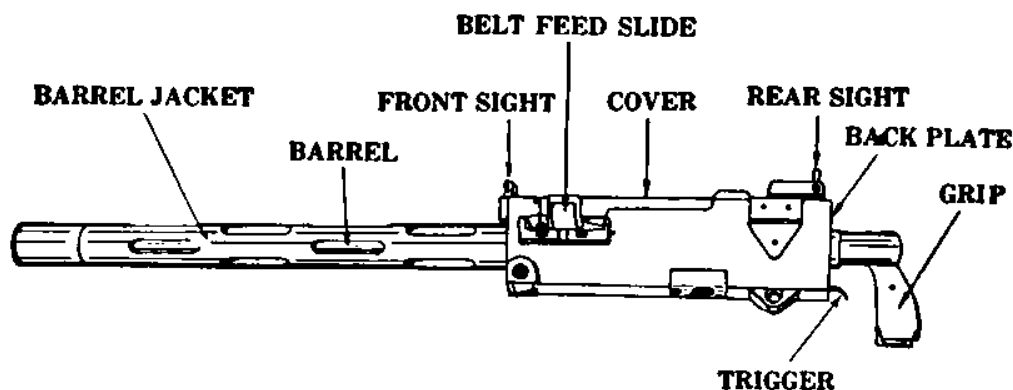


FIGURE 1.—Browning machine gun, M1919A4.

■ **2. COOLING SYSTEM.**—The machine gun, caliber .30, M1919A4, is provided with a heavy barrel which is exposed to the air. This factor serves to keep the gun at operating temperatures under normal conditions which is at the rate of about 60 rounds per minute for about 30 minutes.

■ **3. AMMUNITION BELT.**—Woven fabric belts of a capacity of 250 rounds (equipped with brass strips at each end to facilitate loading the gun) are normally used with the ground light machine gun.

■ 4. MOUNTINGS.—*a.* The ground light machine gun normally is mounted on the light machine gun tripod M2, a description of which is given in section VI.

b. In motorized or mechanized units, the light machine gun is mounted on vehicular mounts of several types, but a light machine-gun tripod M2 is usually carried for each gun so that it may be fired from the ground.

■ 5. GENERAL DATA.—General data for the light machine gun and mount are as follows:

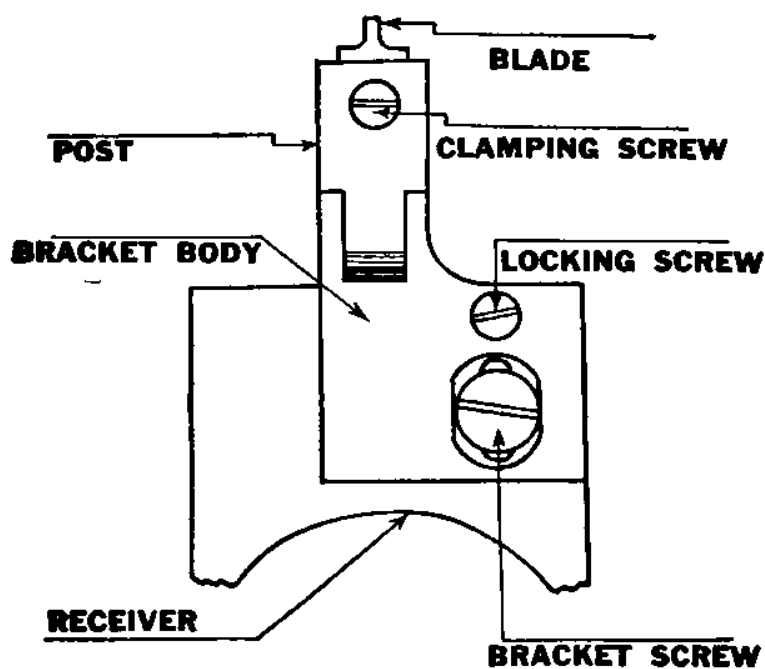
Weight of gun-----pounds--	31. 50
Weight of gun (with pintle and combined elevating and traversing mechanism, approximate weight 4.75 pounds)-----pounds--	36. 25
Weight of gun and tripod mount-----do--	48. 00
Weight of tripod mount M2-----do--	16. 50
Weight of tripod (less pintle and elevating-traversing mechanism)-----pounds--	11. 75
Length (over-all)-----inches--	41. 11
Length of barrel-----do--	24.
Caliber -----do--	. 30
Rifling:	
Number of grooves-----	4
R. H. twist, 1 turn in 33.3 cal. or--inches--	10.
Depth of grooves-----do--	.004
Trigger pull-----pounds--	7. 7
Sights graduated to-----yards--	2, 400
Cyclic rate of fire----- rounds per minute--	450-625
Maximum usable rate of fire-----do--	150
Muzzle velocity, M2 ball cartridge	
feet per second--	2, 800
Muzzle velocity, M1 ball cartridge-----do--	2, 600
Capacity of ammunition chest, wooden	
rounds--	250
Capacity of ammunition chest M1-----do--	300
Weight of ammunition chest, wooden (empty)	
pounds--	5. 5
Weight of ammunition chest M1 (empty)	
do----	2. 43

Weight of spare parts and accessory chest M5 (with contents) -----pounds--	18. 93
Characteristics of tripod mount M2:	
Length extended-----inches--	32. 5
Length folded for transportation__do----	27.
Maximum search:	
With elevating mechanism attached:	
Above zero graduation----mils--	+265
Below zero graduation-----do--	-200
(Elevating handwheel gradu- ated every mil.)	
Without elevating mechanism at- tached: Variable, but in excess of mil figures given above.	
Maximum traverse:	
With traversing handwheel---mils--	61
Additional on traversing bar:	
Right from zero graduation mils----	450
Left from zero graduation mils----	425
As a free gun-----do----	6, 400

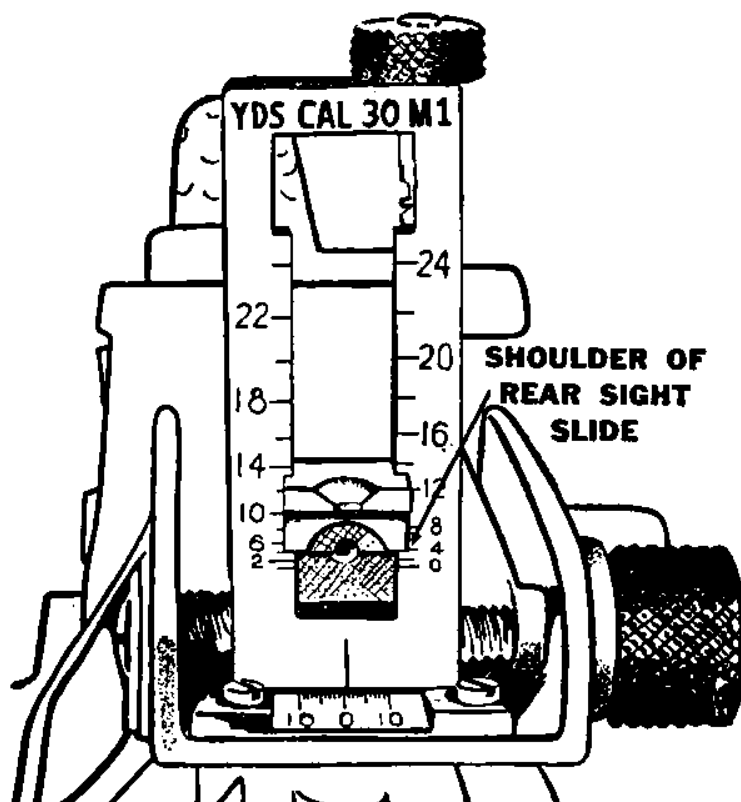
NOTE.—The above figures on search and traverse may vary slightly for different sets of equipment.

■ **6. SIGHTS.**—*a. Front.*—The front sight consists of a front sight blade, a front sight body, a front sight post, and a plunger mechanism. The front sight post pivots on the front sight bearing screw when folded for convenience in packing. The plunger mechanism provides a locking device to keep the front sight post in its upright position when the gun is being fired. The front sight is attached to the front end of the receiver by means of a screw. The height of the front sight is such that when the rear sight slide is set at an elevation a bullet fired from the gun will strike a target at a distance corresponding to the elevation set on the rear sight.

b. Rear.—(1) The rear sight (fig. 2) is of conventional type. It consists of a rear sight leaf, carrying a peep in the slide mounting, pivoted on the rear sight base, and adjustable for windage. The rear sight base mounts the rear sight



① Front sight.



② Rear sight.

FIGURE 2.—Sights.

leaf and rear sight leaf spring. It is secured to the left side plate of the receiver by three screws in the flange of the base.

(2) The rear sight leaf is graduated for elevation in 100-yard divisions up to 2,400 yards. The peep of the rear sight slide is 0.081 inch in diameter. Motion of the rear sight slide is accomplished by rotation of the elevating screw knob. This elevating screw mechanism is equipped with a mil click device which may be used in conjunction with a mil scale engraved on the left side of the rear sight leaf to measure or establish angles of elevation in mils.

(3) The windage screw mechanism also incorporates a mil click device. Adjustment of the rear sight leaf in windage is accomplished by rotation of the windage screw knob. Amount of motion permitted is 10 mils right or left from zero.

(4) The sight radius is 13.94 inches.

■ **7. PINTLE.**—The pintle of the light machine gun (ground), although technically not a part of the gun, is permanently assembled thereto by a bolt through the trunnions of the pintle and the trunnion hole of the receiver of the gun. Failure to keep this bolt reasonably tight will produce inaccurate fire. This pintle is tapered and mates with the corresponding tapered pintle bushing of the head of the tripod mount M2. This tapered pintle thus serves as a tight bearing union between the receiver of the gun and its mounting. The pintle is secured in its mounting by the engagement of a spring actuated pintle latch of the mounting in a corresponding annular groove of the pintle.

■ **8. ELEVATING AND TRAVERSING MECHANISM.**—*a.* As with the pintle, the elevating and traversing mechanism is not technically a part of the gun. However, the elevating and traversing mechanism is permanently secured to the receiver of the gun by a bolt through the head of this mechanism and the elevating bracket of the gun. In guns of new manufacture, the elevating bracket is integral with the bottom plate.

b. The elevating and traversing mechanism, when used with the tripod M2 consists of an upper elevating screw; a

lower elevating screw; an elevating handwheel assembly secured to the head of the lower elevating screw; a housing mating with the lower elevating screw; a traversing slide mounted to the lower elevating screw housing. The new type traversing mechanism also includes a traversing screw, handwheel, and micrometer. This mechanism incorporates a mil click device.

(1) The upper elevating screw terminates at its upper end in an offset head which incorporates a recess for the bolt which assembles the entire elevating and traversing mechanism to the gun. The mechanism is properly assembled to the gun when the offset head points to the rear, thus permitting the mechanism to be folded to the rear and seated in its recess in the duralumin grip. The upper elevating screw is externally threaded to mate with the internal threads of the lower elevating screw. It is equipped with a longitudinal slot in which is seated an engraved scale. This scale is utilized to indicate plus or minus increments of elevation given the mounted gun. It is subdivided into 50-mil graduations and is read by noting the position of the upper edge of the lug of the click ring which moves in the longitudinal slot as the elevating handwheel is rotated.

(2) The lower elevating screw is threaded internally to mate with the externally threaded upper elevating screw and is threaded externally to mate with the lower elevating screw housing. Secured to its upper end is the elevating handwheel assembly.

(3) The elevating handwheel assembly incorporates a mil click device. The handwheel is of light duralumin alloy and carries around its outer periphery ten notches, each notch indicating a 5-mil increment of elevation. Engraved on the upper surface of the handwheel is a mil scale which is read directly from the indicator attached to the click ring. This scale is of 50-mil amplitude and is divided and numbered in 5-mil major divisions and single-mil subdivisions. The click ring which is carried in the elevating handwheel does not rotate with the handwheel, being prevented therefrom by engagement of a lug in the longitudinal slot of the upper elevating screw.

(4) The lower elevating screw sleeve is threaded internally to engage the external threads of the lower elevating screw. This sleeve carries at its front portion a spring actuated plunger which serves to prevent disassembly of the lower elevating screw from its sleeve. At the upper rear portion of the lower elevating screw sleeve is the male portion of a swivel joint, attaching the traversing slide to the elevating and traversing mechanism.

(5) The traversing slide mounts the traversing slide lock lever assembly. This assembly consists of the traversing slide lock lever, the traversing slide lock lever screw, the traversing slide lock lever positioning the traversing slide lock spring, the traversing slide locking screw and washer. Adjustment of the lever to the clamping screw is provided for by a spline assembly, in turn secured by a locking screw and washer. Adjustment is correct when the clamping screw is in firm contact with the traversing bar, the lever being about 45° above the right horizontal position.

(6) Movement of the gun in elevation, when mounted on the tripod M2 is accomplished by rotation of the elevating handwheel, the traversing slide being in firm contact with the traversing bar, with the traversing slide lock lever preferably engaged. (When firing on rapidly moving ground targets, the traversing slide lock is not engaged, although the traversing slide must be retained in firm contact with the traversing bar.)

(7) When mounted on the M2 tripod, the light machine gun may be traversed—

(a) When equipped with the old type traversing mechanism, by moving the gun right or left as desired, the traversing clamp being disengaged from the traversing bar, while the traversing slide is retained in firm contact with the traversing bar.

(b) When equipped with the new type mechanism: beyond 50 mils, as stated above; under 50 mils, by using the traversing hand wheel.

SECTION II

DISASSEMBLING, ASSEMBLING, AND CHANGING
PARTS

■ 9. GENERAL.—*a.* Disassembling may be considered under two general headings: first, removal of groups to the extent required for ordinary cleaning and minor repairs; and, second, detailed disassembling, involving removal of all components from each group.

b. A group is a number of parts, contained in a common housing, which function as a unit.

c. The removal of the different groups from the gun and complete disassembling of the groups to be disassembled by the using services can be accomplished with the tools provided.

■ 10. REMOVAL OF GROUPS FROM GUN.—*a. Backplate.*—(1) Pull back on latch and raise cover. With the left hand pull back bolt handle and hold it in the rearmost position.

(2) Insert rim of a cartridge in slit in end of driving spring rod. With slit horizontal, push in driving spring rod as far as it will go and turn it clockwise one-quarter turn until slit is vertical. In this position the lugs on it will engage in their recesses in the bolt.

(3) Push bolt handle forward about an inch to free the rear end of driving spring rod from backplate.

(4) Push latch forward and lift out backplate.

b. Bolt handle.—Pull bolt all the way back and remove bolt handle.

c. Bolt.—Remove bolt from rear end of receiver being careful not to handle driving spring rod.

d. Lock frame.—Insert nose of cartridge through hole in the right side of receiver and push in on trigger pin. Grasp trigger and pull lock frame, barrel extension, and barrel out of receiver. Hold barrel in one hand, lock frame in the other, and push forward on accelerator. This separates lock frame and barrel extension.

e. Barrel extension and barrel.—Unscrew barrel extension from barrel.

f. Latch.—With the left hand, palm up, against the rear of the receiver to prevent dropping the latch spring, pull latch to rear until it separates from the plate. Guns of recent manufacture have the latch spring riveted to the latch.

g. Latch spring.—If not riveted lift the latch spring from its pin in the latch.

h. Cover.—(1) *Guns of early manufacture.*—(a) Turn cover pin spring up and remove pin.

(b) Remove cover.

(2) *Guns of new manufacture.*—(a) Remove cotter pin from cover bolt.

(b) Place screwdriver blade of combination tool in slot in cover bolt to prevent it from turning and with a wrench remove the nut.

(c) Remove cover bolt, cover catch spring, and fixed and movable plates.

(d) Remove cover.

NOTE.—To prevent undue wear, the cover and latch should not be removed except when necessary for cleaning or replacement of parts.

■ **11. REPLACING GROUPS IN GUN.**—In general, the groups are replaced in the gun in reverse order.

a. (1) Guns of early manufacture.—Replace cover. Insert cover pin and lock it by turning cover pin spring forward into its seat in trunnion block. Some models have a cover pin which is locked by inserting and spreading a cotter pin through a hole in the cover pin.

(2) *Guns of new manufacture.*—Replace cover. Place cover latch spring on bolt and position the fixed and movable plates. Insert cover bolt into cover hole. Place screwdriver blade of combination tool in slot in cover bolt to prevent it from turning, and assemble nut to cover bolt using a wrench until the desired tension is obtained. Replace cotter pin.

b. Seat rounded end of latch spring in latch seat, placing hole in spring over pin, bent side of spring away from latch. Holding spring in place with the fingers, push latch onto top plate from the rear, free end of the spring to the front, then force the latch home. If the spring is allowed to slip from its seat, the latch will not function, and the spring will jam the latch so that it cannot be removed without breaking.

c. Screw barrel into barrel extension until the rear end of the barrel is flush with the inside of the barrel extension.

d. Insert barrel and barrel extension into receiver until the forward end of barrel extension is opposite the rear end of receiver.

e. Holding barrel extension with one hand, take lock frame in other hand, with index finger beneath and supporting accelerator. Place claws of accelerator in front of and against T-lug. Insert front projections of lock frame into slots of barrel extension and push forward until accelerator turns backward, locking lock frame to barrel extension. Push down tips of accelerator to insure positive locking.

f. Push parts into gun, forcing trigger pin inward to clear it from right side plate, and push forward until a click is heard as trigger pin springs out into its seat in the right side plate. (If barrel hangs on front barrel bearing, reach forward under jacket and align it.)

g. Push cocking lever forward and insert bolt, pushing down on rear end of trigger to prevent ejector from tripping accelerator.

h. Insert bolt handle through large opening at rear of slot and push it forward about 1 inch, being sure that collar on handle is inside right side plate.

i. Push forward on latch and replace backplate.

j. Hold bolt handle fully back with left hand. Place rim of cartridge in slit in end of driving spring rod, and turn rod one-quarter turn counterclockwise until slit is horizontal. This releases driving spring. Allow bolt to go forward.

k. Make head space adjustment as follows:

(1) Pull bolt to rear about $\frac{3}{4}$ inch.

(2) Screw barrel into barrel extension (by using point of a cartridge or the combination tool in barrel notches) until the action will *just close* (recoiling parts will go fully forward) without being forced.

(3) Unscrew the barrel one notch.

Caution: Care must be exercised to avoid roughening the barrel surface during the adjustment.

l. Position belt feed lever stud over cam groove in bolt and close cover.

m. Pull trigger.

NOTES.—1. HEAD SPACE.—*a.* The head space of a military weapon with a cartridge fully seated in the chamber is the distance between the base of the cartridge and the face of the bolt, when the bolt is in its locked position.

b. In Browning machine guns, the head space is adjusted by obtaining the proper distance between the forward part of the bolt and the rear end of the barrel. The head space adjustment must be checked before firing.

2. EFFECT OF HEAD SPACE ADJUSTMENT.—*a. General.*—The head space adjustment of the machine gun is the most important adjustment to be made. To insure continued operation of the gun, to obtain the best uniformity of shot patterns, and to prevent damage to the gun, the head space adjustment should be made as prescribed in paragraph 11*k* above, or note 4 below.

b. Tight head space (fig. 3).—When the head space adjustment is tight, poor functioning will result, as the breech lock will not fully enter its recess in the bolt. This condition may damage the barrel extension, bolt, or breech lock. Extraction trouble may also occur because of improper timing of locking and unlocking. Furthermore, with a tight head space adjustment the gun operates sluggishly because of the binding of the moving parts. If the head space adjustment is very tight, the notches on the rear end of the barrel can be seen, the bolt handle will not be fully forward, and the firing pin cannot be released by pressing the trigger.

c. Loose head space (fig. 4).—If the head space is loose, a ruptured cartridge may occur; also if there is a weakness in the cartridge case itself, such as a split case, the possibility of a ruptured cartridge is increased. If the head space adjustment is loose, the bolt can be moved slightly from front to rear independently of the barrel and barrel extension.

3. QUICK HEAD SPACE ADJUSTMENT.—After the head space adjustment has been determined by the method described in paragraph 11*k*, or by the alternate method described in paragraph 4 below, the notch in which the barrel locking spring is engaged may be marked with a center punch. Thereafter, to make the correct head space adjustment during the assembly of the gun, screw the barrel all of the way into the barrel extension and then unscrew the barrel until the barrel locking spring is in the marked notch.

4. ALTERNATE HEAD SPACE ADJUSTMENT.—Head space adjustment can be made before the moving parts are assembled in the receiver. Screw the barrel into the barrel extension until the rear of the barrel is flush with the inside of the barrel extension. Remove the extractor from the bolt. Place the bolt into the bolt guides in the barrel extension. Push the bolt fully forward. Turn the assembly so that the bolt is down. Lock the bolt to the barrel extension by pushing the breech lock fully into its seat in the bolt. Hold it firmly in that position with the thumb. Screw the barrel into the barrel extension until it is stopped by contact with the forward face of the bolt. Be certain that the barrel does not force the breech lock out of its fully locked position. Turn the assembly so that the bolt is up. Unscrew the barrel from the barrel extension just enough to cause the breech lock to fall from its recess in the bolt. Remove the bolt. Screw the barrel into the barrel extension one notch, or, if the nose of the barrel locking spring is between two notches, screw the barrel into the barrel extension one and the fraction notches. Assemble the gun.

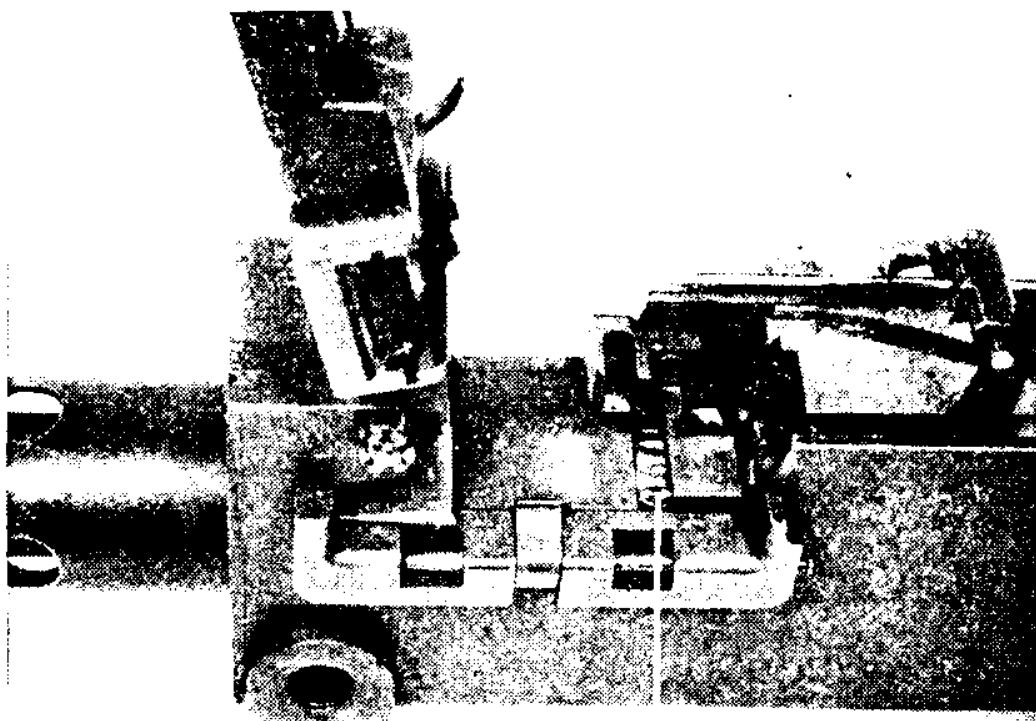


FIGURE 3.—Tight head space.

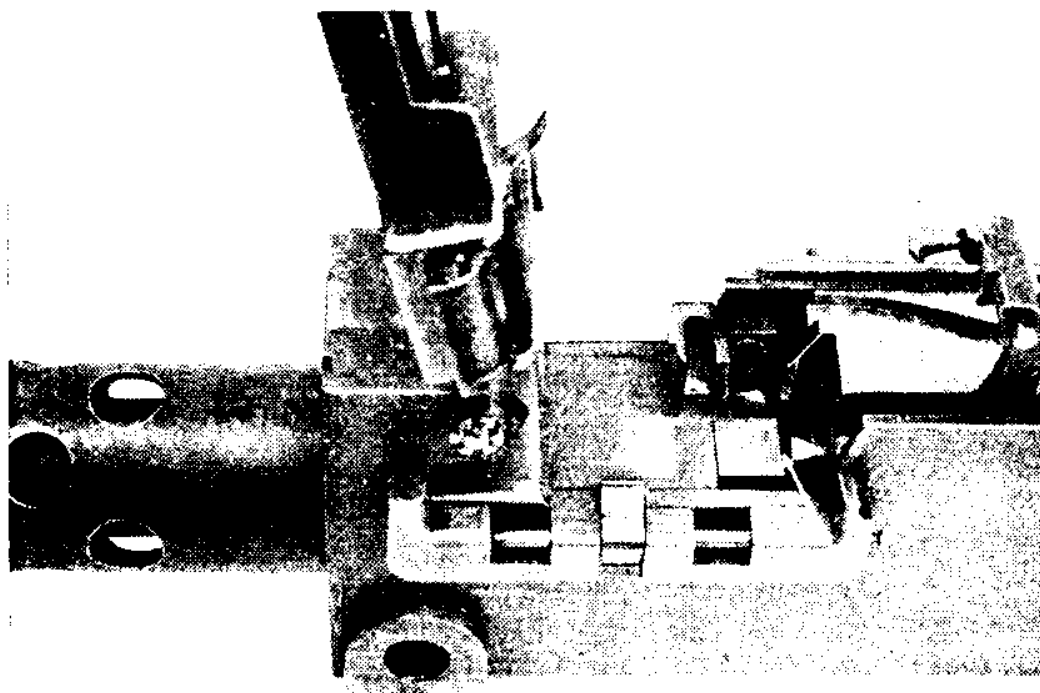


FIGURE 4.—Loose head space.

■ 12. DETAILED DISASSEMBLING OF BOLT.—*a.* Turn the extractor up and remove it to the left (fig. 5).

b. Great care should be exercised in removing the driving spring rod from the bolt as the force of the driving spring when released can easily cause the rod to slip away from the hand and possibly result in serious injury. To remove driving spring rod, place protruding end of rod on the table or a block of wood. With bolt firmly grasped by the right hand (palm of the hand over face of bolt), press down and at the

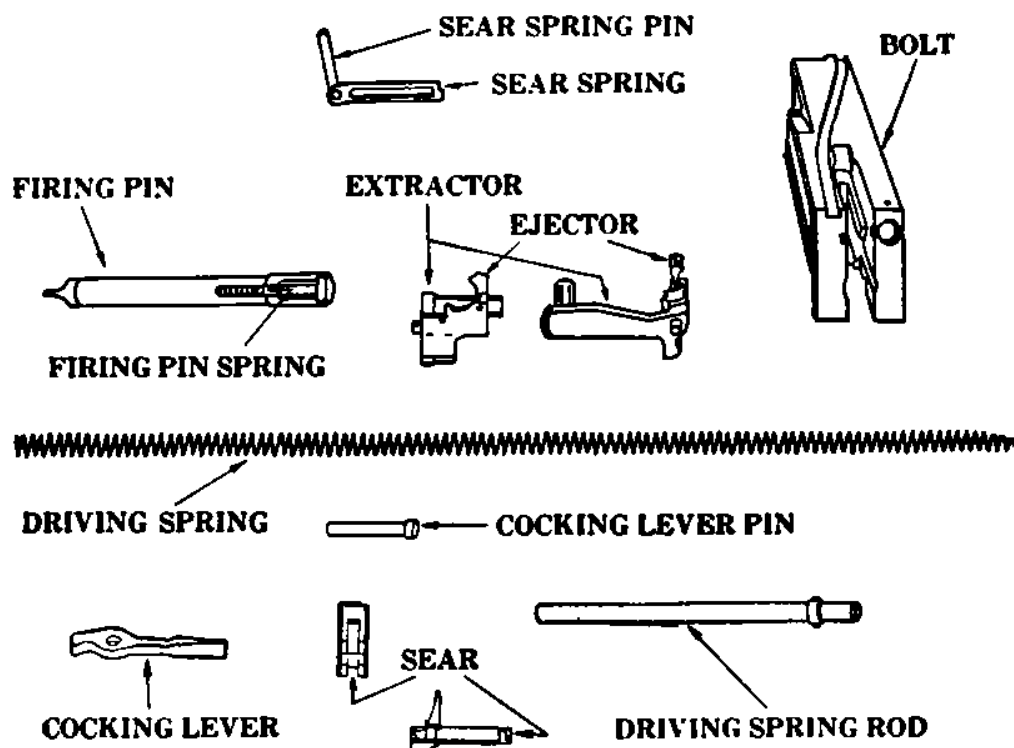


FIGURE 5.—Bolt group.

same time turn bolt one-fourth turn to the left until lugs on rod leave their recess in bolt. Slowly release pressure on bolt, allowing it to rise under the action of driving spring until about 3 inches of rod protrude. With the left hand grasp protruding portion of rod and spring; raise both hands and the bolt from the table, keeping rod and spring in their same position relative to bolt. Separate rod and spring from bolt with a quick jerk. The quick, separating jerk will not allow spring to kink. Separate driving spring rod and driving spring.

c. Turn the top of cocking lever to the rear of bolt and withdraw cocking lever pin to the left of bolt.

d. Lift out cocking lever.

e. Release firing pin by pushing down on sear. Hold bolt in the left hand, the front end toward the body, top up, with the index finger of the left hand beneath and supporting the sear. Use the nose of a cartridge, placed near the end of sear spring, to push downward and to the right on spring to seat it in cut in bolt. This releases sear which is removed at the bottom of bolt.

f. Turn sear spring back to the left to clear the cut. Push nose of a cartridge into the hole in the bottom of bolt to start sear spring pin moving. To complete the removing of sear spring, place the top end of cocking lever well under sear spring and pry down against the edge of bolt.

g. Place the palm of the right hand over the rear of bolt, tilt the rear end of bolt down, and firing pin will drop out.

■ **13. ASSEMBLING BOLT.**—a. Place firing pin in bolt, striker downward and to the front, and tilt the front of bolt downward until striker projects through the small hole in the front of bolt.

b. Replace sear spring by pushing with a cartridge on top of pin, avoiding pressure on spring proper.

c. Hold bolt in the left hand, front end toward the body, top up. With point of a cartridge placed near the end of sear spring, push downward and to the right to seat it in cut in bolt.

d. Push sear upward from the bottom, notched projection toward the front of bolt, and hold with first finger of the left hand while pressing downward and to the left on sear spring with a cartridge to engage the end of sear spring in sear.

e. Replace cocking lever, making certain that the rounded nose on the lower end is to the rear of bolt so that it will properly engage in the recess in firing pin.

f. Insert cocking lever pin on the left side of the bolt. The upper end of the cocking lever should be to the rear of the bolt before inserting the pin. This alines the hole in the

cocking lever with the hole on either side of the bolt for the return of the cocking lever pin.

g. Cock by pressing forward on cocking lever. Turn cocking lever to the rear and press down on sear with a cartridge to release firing pin and test the correctness of the assembly. Recock the assembly.

h. The same care should be exercised in assembling the driving spring rod to the bolt that is exercised in removing it. Place driving spring on driving spring rod. With the back end of the rod resting on a table or a block of wood, gather as much of the spring on the rod as can be held compressed by the thumb and fingers of the left hand. With bolt securely held in the right hand, the front end of bolt in the palm of the hand, slip bolt over the end of spring. Push downward to compress spring and allow lugs on rod to enter

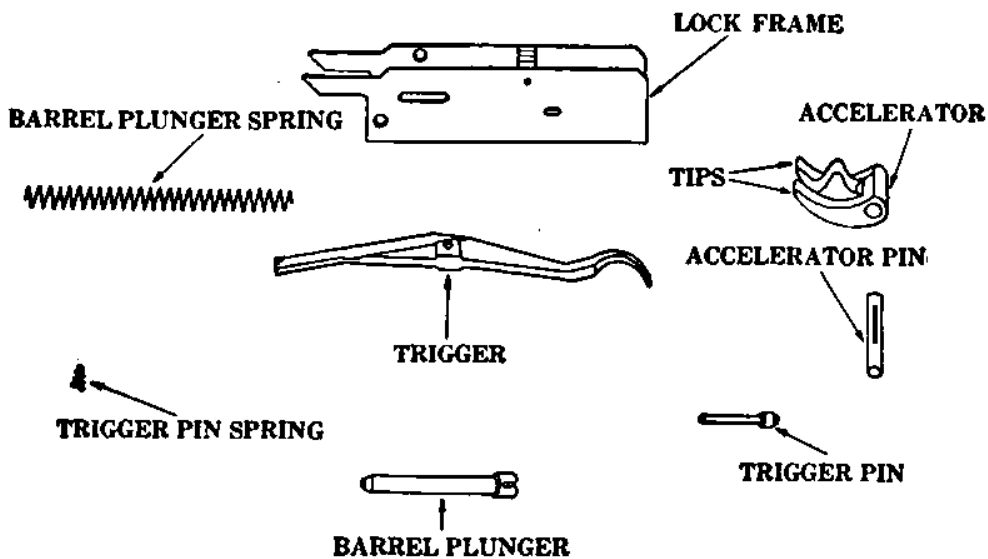


FIGURE 6.—Lock frame group.

slot in bolt. Turn bolt slowly 90° clockwise until slit in rod is crosswise to slot in bolt.

i. Insert pin on extractor into the rear one of the two large holes in the left side of bolt, extractor pointing up. Turn extractor downward toward the front to engage collar on extractor under collar cut in bolt.

■ 14. DETAILED DISASSEMBLING OF LOCK FRAME (fig. 6).—*a.* Grasp head of trigger pin between the thumb and first finger of the right hand and remove it to the right. Lift out trigger. If pin is too tight to permit its removal in this manner,

it must be drifted out. Do not remove trigger pin spring except when necessary.

b. Push out accelerator pin and remove accelerator.

c. Hold lock frame with the left hand, projections pointing upward, slot to the left, separator between the second and third fingers, first and second fingers gripping barrel plunger spring. With the thumb of the right hand, press down and out on barrel plunger to disengage plunger guide pin from slot. Allow spring, with plunger, to rise slowly. Lift out spring and remove it from barrel plunger.

■ 15. ASSEMBLING LOCK FRAME.—a. Assemble barrel plunger spring to barrel plunger, being careful that the more tightly fitting end of barrel plunger spring is pushed up against the head of barrel plunger. Hold lock frame with the left hand, projections pointing upward, slot to the left, lock frame separator between the second and third fingers. Seat the end of barrel plunger spring in the recess in lock frame separator, barrel plunger guide pin facing the slot in lock frame. Using the first and second fingers of the left hand to prevent spring from buckling, press down with the thumb of the right hand on the end of barrel plunger until barrel plunger guide pin can be seated in the slot. Care should be taken that the action of the spring does not cause the plunger to slip out of the hand.

b. Replace accelerator with the tips up and the rounded surface to the front. Insert accelerator pin, taking care that both ends of pin are flush with the sides of lock frame.

c. Push the front end of trigger up between separator and spacer, placing the center in its square seating. If trigger pin spring has been removed, seat spring on trigger pin, placing the small end of spring toward the head of pin. Replace pin from the right.

■ 16. DETAILED DISASSEMBLING OF BARREL EXTENSION (fig. 7).—a. Insert the rim of a cartridge under the front edge of barrel locking spring and pull it out to the front.

b. Push out breech lock pin and remove breech lock.

■ 17. ASSEMBLING BARREL EXTENSION.—a. Place breech lock in its slot, *taking care that the double beveled surface is up*

and to the front. Insert breech lock pin and insure that both ends of pin are flush with the sides of barrel extension.

b. Insert barrel locking spring in the seating in the left side of barrel extension, hook inward, and force home as far as it will go.

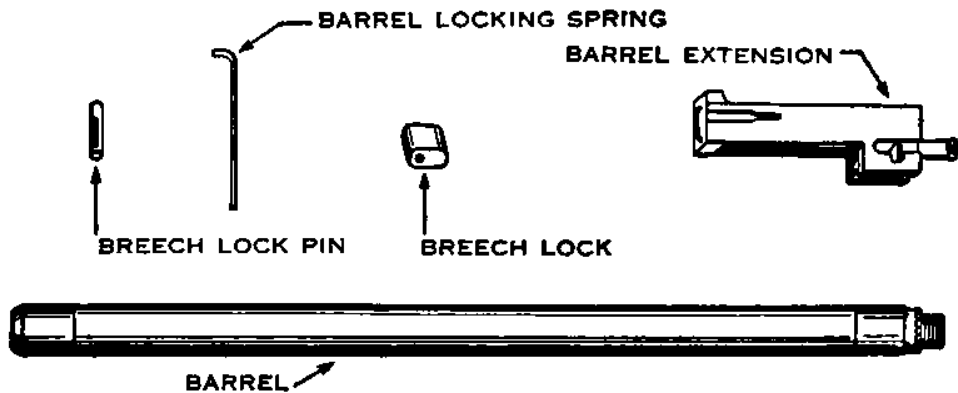


FIGURE 7.—Barrel and barrel extension group.

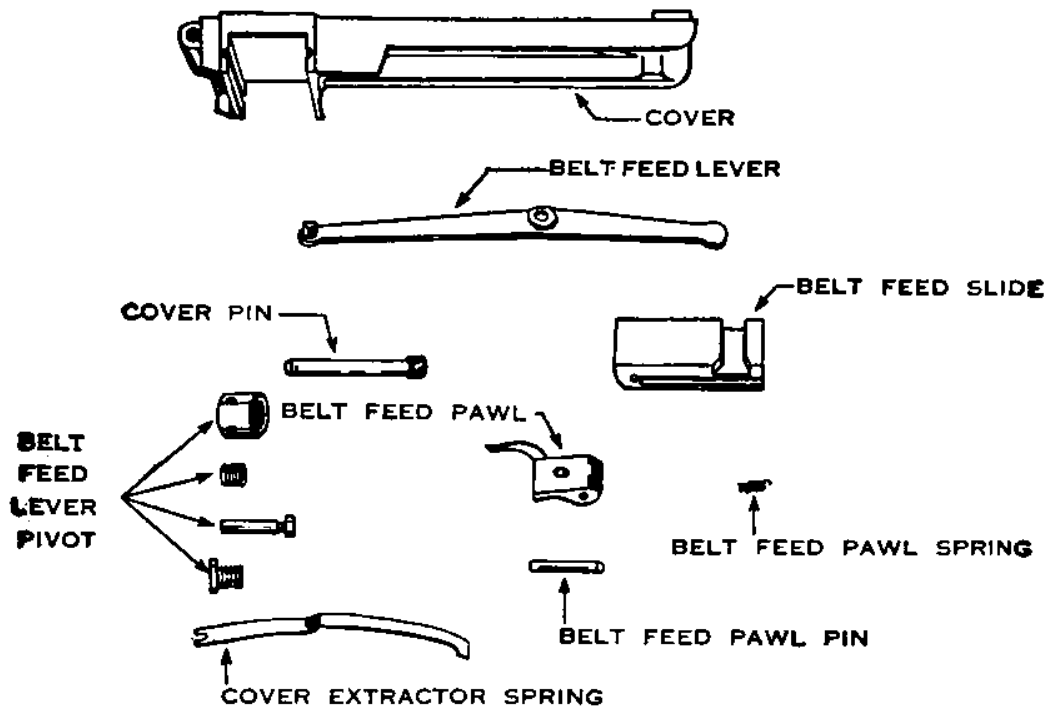


FIGURE 8.—Feed group.

■ 18. DETAILED DISASSEMBLING OF COVER (fig. 8).—a. Remove cap or plug from belt feed lever pivot bushing nut. Remove belt feed lever pivot.

b. Withdraw belt feed lever from belt feed slide and remove slide.

c. Insert the nose of a cartridge between cover extractor spring and the notch in cover extractor cam. With the thumb of the left hand over spring, pry out on spring to disengage from cut. Lift out spring from its seat against stud.

■ 19. ASSEMBLING COVER.—*a.* Place the forked end of cover extractor spring under stud on cover. Press downward with the thumb on the other end of spring, at the same time pushing toward stud, and seat projection of spring in the notch of cover extractor cam.

b. Replace belt feed slide in its grooves in cover, taking care that pawl is pointing to the right as cover goes on the gun.

c. Place front end of belt feed lever in the cut, stud on lever away from cover and to the rear. Insert belt feed lever pivot pin. Replace cotter pin or cap.

■ 20. DISASSEMBLING AND ASSEMBLING OF PARTS DISMOUNTED ONLY FOR REPAIR.—*a. Shock-absorbing group.*—(1) *Disassembly.*—Unscrew the adjusting screw; remove the adjusting screw plunger and screw.

(a) *Disk type.*—Remove the buffer disks, buffer plug, buffer ring, and the buffer plate through the rear end of the grip.

(b) *Spring type.*—Remove the buffer disk, stop, spring, filler, and buffer plate.

(2) *Assembly.*—(a) *Disk type.*—In assembling the shock-absorbing group, it must be kept in mind that part of the recoil energy of the bolt is transmitted to the buffer pile in the form of metal to metal impact of approximately 5,600 pounds. This force is absorbed by the fiber buffer disks. In order that the buffer pile may perform this function properly, the following precautions must be observed:

1. The disks must form a compact column or stack from the face of the buffer plate to the adjusting screw.
2. The disks should be clean and free from rough edges and surfaces, and should be assembled in the tube one at a time. Be sure that each disk

is firmly seated and use sufficient disks (normally 16) so that when the adjusting screw is inserted and tightened its rear face will be flush with or protrude slightly from the rear face of the grip.

3. After assembly, adjust the buffer by tightening the adjusting screw until the buffer plate can just be turned by grasping it by the end which extends through the backplate. When the gun is assembled, this adjustment can be made approximately without removing the backplate by tightening the adjusting screw as far as it will go and then loosening one-half turn.

4. These disks are used for their shock-absorbing capacity, and substitution of other material for or removal of the disks is prohibited (par. 247b(2)).

(b) *Spring type*.—Assemble in reverse order. Tighten the adjusting screw as far as possible with the combination tool and then loosen $1\frac{1}{2}$ turns. For additional data on assembling groups see paragraph 239.

b. *Belt holding pawl*.—(1) Hold down belt holding pawl and withdraw belt holding pawl split pin to the rear.

(2) Lift off belt holding pawl.

(3) Lift belt holding pawl spring from its seating.

(4) Replace in the reverse order.

■ 21. **CHANGING PARTS**.—If the time element is important, a broken minor part should be replaced by substituting a complete spare group which contains it. Thus a broken firing pin would be remedied by changing bolts. Replacement parts in the spare parts chest should be replaced as soon as opportunity permits. In the event that either the bolt, barrel, or barrel extension is changed, *readjust the head space*.

SECTION III

CARE AND CLEANING

■ 22. **IMPORTANCE**.—The care and cleaning of the machine gun, tripod, and accessories are a vital duty. Experience has shown that most machine guns become unserviceable from lack of proper care rather than from use.

■ 23. LUBRICANTS, CLEANING MATERIALS, AND RUST PREVENTIVES.—*a. General.*—The use of unauthorized materials, such as abrasives, is forbidden. The following are the only materials authorized and issued for cleaning these machine guns:

Cleaner, rifle bore.

Soda ash.

Soap.

Oil, lubricating, preservative, light.

Oil, lubricating for aircraft instruments and machine guns.

Compound, rust-preventive, light.

Solvent, dry-cleaning.

Decontaminating agents.

Water.

b. Rifle bore cleaner.—(1) Rifle bore cleaner is issued for cleaning the bore of the machine gun after firing. This material possesses rust-preventive properties and provides temporary protection against rust. It is better practice, however, to dry the bore immediately after cleaning it with rifle bore cleaner and then coat it lightly with light preservative lubricating oil.

(2) Rifle bore cleaner freezes at temperatures below 32° F. If frozen, it must be thawed and shaken well before using. To prevent bursting, containers should not be filled more than three-fourths full in freezing weather.

c. Soda ash.—Soda ash is a white, odorless powder which is soluble in water. For use, it is dissolved in water in the proportion of $\frac{1}{2}$ to 1 pound of soda ash to 1 gallon of boiling water. If boiling water is not available, hot or cold water may be used. This solution can be used to clean the gun bore when rifle bore cleaner is not available.

d. Soap.—A soap solution may be used to clean the bore and mechanism, if neither rifle bore cleaner nor soda ash is available. The solution is prepared by dissolving soap chips in hot water in the proportion of $\frac{1}{4}$ pound to 1 gallon of water. When possible, the solution should be used while it is hot.

e. Light preservative lubricating oil.—This oil has both rust-preventive and lubricating properties but cannot be

depended upon to provide protection from rust for long periods. It is used for the lubrication of all moving parts and for *short term* protection against rust. Preservative action results partly from the oily film on the metal parts and partly from the chemical combination of inhibitors in the oil with the metal. It will protect the metal surfaces from rust even though no appreciable film of oil is present on the metal parts. When used on moving parts, however, it is necessary to maintain a thin film of oil to provide the necessary lubrication.

f. Lubricating oil for aircraft instruments and machine guns.—This oil, which is a lubricant rather than a rust-preventive agent, may be used for lubricating the machine gun when light preservative lubricating oil is not available. It is an extremely light oil which relies entirely upon the maintenance of the oil film to protect metal surfaces from rusting. *When it is used as a preservative, the parts must be inspected daily for rust.* Following this inspection, the metal parts should be wiped clean and again lightly coated with the oil.

g. Light rust-preventive compound.—This compound is issued for the protection of metal parts for long-time storage while the parts are boxed. It is a sluggish liquid at about 60° F. It can usually be applied with a brush.

h. Dry-cleaning solvent.—This is a noncorrosive, petroleum solvent which is used to remove grease, oil, or rust-preventive compound. *Dry-cleaning solvent is highly inflammable and should not be used near open flames. Smoking is prohibited where dry-cleaning solvent is used.* The solvent is generally applied with rag swabs to large parts of the gun and used as a bath for the small parts. All surfaces must be thoroughly dried with clean rags immediately after use of the solvent. Gloves should be worn when handling parts that have been cleaned, in order to avoid leaving finger marks which are usually acid and therefore induce corrosion. Cleaning solvent attacks and discolors rubber.

■ 24. CARE AND CLEANING WHEN NO FIRING IS DONE.—*a. General.*—(1) This includes the care necessary to preserve the

condition of the machine gun and mount during the time when no firing is being done.

(2) The gun must be completely disassembled for proper inspection, cleaning, and lubrication.

(3) Machine guns in the hands of troops should be inspected daily during periods of use to insure proper condition and cleanliness. Training schedules should allow time for supervised cleaning on each day that guns are used.

b. Bore and chamber.—(1) To clean the bore and chamber, assemble a cloth patch to the cleaning rod and insert the rod into the bore *at the breech end*. Move it forward and backward several times and replace it with a fresh new patch. Make sure that the patch goes all the way through the bore before the direction is reversed. This cleaning removes accumulations of dust, dirt, and thickened oil. Be sure that the chamber as well as the bore is thoroughly cleaned.

(2) Continue this operation until a patch comes out clean.

(3) After thoroughly cleaning, saturate a patch with light preservative lubricating oil and push it through the chamber and bore.

(4) When issue patches are not available, patches should be cut approximately 2½ inches square to permit their passage through the bore without bending the cleaning rod.

c. Receiver.—(1) Disassemble, clean, and oil all moving parts of the receiver.

(2) Clean the screw heads and crevices with a small cleaning brush or small stick.

(3) Clean moisture and dirt from the metal surfaces with a dry cloth, then wipe them with a cloth that has been dipped in light preservative lubricating oil. This protective film must be maintained at all times.

(4) Clean the dirt from outer surfaces of the machine gun with a slightly oiled cloth and then wipe with a soft dry one.

d. Accessories.—(1) The mount, spare parts, and all accessories of the gun must be cared for and cleaned as carefully as the gun itself.

(2) Ammunition must be kept clean and dry.

(3) Guns mounted in vehicles are usually locked in traveling position.

(4) Muzzle covers are provided to prevent dirt from entering the gun through the muzzle.

■ 25. PREPARATORY TO FIRING.—The following steps are taken before firing to insure efficient functioning of the gun:

- a. Dismount the main groups.
- b. Clean the bore and chamber with a clean dry patch. *Do not oil the bore and chamber before firing.*
- c. Thoroughly clean and lightly oil all other metal parts with light preservative lubricating oil. Do not use grease. Be sure to apply a thin coating of oil to the following:
 - (1) All parts of bolt group.
 - (2) All parts of the lock frame group.
 - (3) All parts of the barrel extension group.
 - (4) All parts of the feed group.
- d. Assemble the machine gun and rub all outer surfaces with a lightly oiled rag to remove dust.
- e. Check head space.
- f. Clean and prepare for firing all parts of tripod and other equipment necessary for firing.

■ 26. AFTER FIRING.—a. General.—(1) The bores of all machine guns must be thoroughly cleaned by the evening of the day on which they are fired. They should be cleaned in the same manner for the next 3 days.

(2) Firing the machine gun causes powder and primer fouling to form in the bore and chamber. This fouling absorbs and retains moisture from the air, thereby causing rust. These deposits can be removed by cleaning with rifle bore cleaner, soda ash solution, soap solution, or water.

b. *Cleaning procedure after firing.*—(1) Unload the gun and replace the belt in the ammunition chest.

(2) Clean the bore (c or d below) and all working parts. If this cannot be done at once, apply oil to prevent rust.

(3) Release the firing pin spring.

(4) At the first opportunity, entirely disassemble the gun, clean, oil, and inspect all parts, and make needed repairs and replacements.

(5) On assembly, check operation with dummy cartridges and release the firing pin spring after making sure that functioning and adjustment are correct.

c. Cleaning the barrel with rifle bore cleaner.—(1) Saturate a clean patch with rifle bore cleaner, attach it to the cleaning rod, insert it at the breech end of the barrel, and push back and forth through the bore of the barrel.

(2) Repeat this operation two or three times with clean patches. Be sure that the patch goes completely through the bore before the direction is reversed.

(3) Repeat this procedure with dry patches until a patch comes out clean and dry.

(4) Examine the bore and chamber carefully for cleanliness. If they are not free of all residue, repeat the cleaning process.

(5) If clean, saturate a patch in light preservative lubricating oil and push it through the chamber and bore.

d. Cleaning the barrel with soda ash or soap solution.—(1) Use soda ash solution if rifle bore cleaner is not available. Use soap solution or plain water if neither rifle bore cleaner nor soda ash is available.

(2) Place the barrel, muzzle down, in a vessel containing soda ash or soap solution. (If these are not available use plain hot or cold water.)

(3) Insert the cleaning rod and flannel patch into the breech end. Move the rod up and down for about 1 minute, pumping the solution in and out of the bore.

(4) Run a brass or bronze wire brush forward and back through the barrel three or four times while the bore is wet. The brush must be pushed entirely through the barrel before it is reversed, otherwise the brush will be ruined.

(5) Using the cleaning rod and a clean patch, pump clean water through the bore as in (3) above to remove the washing material.

(6) Remove the barrel from the water and dry it with clean flannel patches.

(7) If the bore and chamber are clean, saturate a patch in light preservative lubricating oil and push it through the bore to oil the bore and chamber.

e. Cleaning parts other than barrel.—(1) Wipe the receiver clean, taking care to remove dirt from all crevices. Thoroughly clean the cover, bolt, barrel extension, lock frame, and backplate, using a small stick covered with a

flannel patch. Remove dirt and oil from all recesses and springs.

(2) Wipe all parts with an oily rag.

f. Front barrel bearing.—(1) Carbon is deposited in the front barrel bearing during firing. If this deposit is not removed periodically, it will eventually cause the barrel to bind.

(2) Unscrew the front plug with the combination tool.

(3) Soften carbon with dry-cleaning solvent.

(4) Scrape out the carbon.

(5) Replace the plug.

g. Exterior surfaces.—(1) Wipe off the exterior of the gun with a dry cloth to remove moisture and dirt.

(2) Wipe all metal surfaces with light preservative lubricating oil.

h. Accessories.—(1) Clean and oil the mount, spare parts, and all accessories with light preservative lubricating oil.

(2) Avoid leaving excess oil in the interior of the tripod head, since this will collect dirt and sand.

(3) The pintle bushing should be cleaned and lightly oiled.

(4) Oil should not be placed in the seating of the traversing dial, since this will collect dirt and interfere with easy dial adjustment.

i. Frequency of cleaning.—Machine guns should be cleaned as soon after firing as possible. *If the gun is not to be fired in the next few days, repeat the procedure outlined above for 3 consecutive days following firing.*

■ 27. ON THE RANGE OR IN THE FIELD.—Machine guns must be kept clean and properly lubricated at all times. To obtain the maximum efficiency of the gun on the range and in the field, the following points must be observed:

a. Never fire a gun with any dust, dirt, mud, or snow in the bore.

b. Keep the chamber free from oil and dirt when firing.

c. Never leave a patch, plug, or other obstruction in the chamber or bore. Failure to observe this precaution may result in serious injury.

d. If the gun gives indication of lack of lubrication and of

excessive friction, apply additional oil to the parts subjected to this friction.

e. Keep a light coating of light preservative lubricating oil on all metal parts.

f. When in the field, the gun must be disassembled to the extent necessary for proper cleaning and lubrication. Upon opportunity, and as necessary, the gun should be disassembled into its main groups, all parts thoroughly cleaned, and the operating parts lubricated before the gun is reassembled.

g. During firing, oil frequently but sparingly those parts where actual friction exists. These include the cam groove and the cocking lever. Excess oil on the moving parts generates smoke which interferes with observation and may disclose the position of the gun.

h. When the prescribed lubricants are not available, any clean, light mineral oil, such as engine oil, may be used. For cleaning the bore and chamber, clean water may be used when no other cleaner is available.

■ 28. PREPARATION FOR STORAGE.—a. Light preservative lubricating oil is the most suitable oil for short-term preservation of the mechanism of the machine gun. It is effective for storage for periods ranging from 2 to 6 weeks, depending on climatic conditions. However, guns which are stored and protected with light preservative lubricating oil must be inspected every 4 or 5 days, and the preservation film renewed if necessary. For longer periods of time, guns must be protected with light rust-preventive compound.

b. Light rust-preventive compound is efficient for preserving the polished surfaces, the bore, and the chamber up to 1 year, depending upon climatic and storage conditions.

c. Machine guns that are to be placed in storage should be cleaned and prepared with particular care. The bore, all parts of the mechanism, and the exterior of the gun should be thoroughly cleaned with dry-cleaning solvent and then dried completely with rags. In damp climates, particular care must be taken to see that the rags are dry. After drying a metal part, the bare hands should not touch that part. All metal parts should then be coated with either light preservative lubricating oil or light rust-preventive compound, de-

pending on the length of storage required. Application of the rust-preventive compound through the bore of the gun is best done by dipping the cleaning brush in the compound and running it entirely through the bore two or three times. The brush must be cleaned before use. Before placing the weapon in storage, the bolt should be in the forward position with the firing pin released. It is good practice to release the firing pin tension whenever the weapon is not in use. The wooden supports of the packing box must be painted with rust-preventive compound before storing the gun. Place the gun in the wooden packing box, handling it with oiled rags. *Under no circumstances will a gun be placed in storage while contained in a cloth or other cover, or with a plug in the bore.* Such articles collect moisture and cause the weapon to rust.

■ 29. **CLEANING WEAPONS RECEIVED FROM STORAGE.**—Weapons which have been stored in accordance with preceding paragraphs will be coated either with light preservative lubricating oil or with light rust-preventive compound. Weapons received from ordnance storage will, in general, be coated with rust-preventive compound. Use dry-cleaning solvent to dissolve and remove all traces of the compound or oil. Take particular care to see that all recesses in which springs or plungers operate are thoroughly cleaned. Failure to do this may cause malfunctioning at normal temperatures and will certainly do so when the rust-preventive compound congeals at low temperatures. After using the cleaning solvent, be sure to dry all parts with dry, clean cloths. Then follow the instructions in paragraph 24.

■ 30. **CARE AND CLEANING UNDER UNUSUAL CONDITIONS.**—*a. Care and cleaning in cold climates.*—(1) In extreme cold, the moving parts of the weapon must be kept absolutely free of moisture. Excess oil or rust-preventive compound on the working parts will solidify to such an extent as to cause sluggish operation or complete failure.

(2) Completely disassemble and clean all parts of the gun thoroughly with dry-cleaning solvent before use in temperature below 0° F. The working surfaces or parts which show

signs of wear may be lubricated by rubbing with a cloth which has been dipped in light preservative lubricating oil and wrung out.

(3) When the gun is brought indoors, it should be allowed to come to room temperature. It should then be disassembled, wiped completely dry of the moisture which will have condensed on the cold metal surfaces, and thoroughly oiled, using light preservative lubricating oil.

(4) If possible, this condensation should be avoided by providing a cold place in which to keep the gun when not in use. For example, a separate cold room with appropriate racks may be used or, when in the field, racks may be set up outdoors under proper cover.

(5) If the gun has been fired, it should be thoroughly cleaned and oiled. The bore may be swabbed out with an oily patch and, after the weapon reaches room temperature, thoroughly cleaned and oiled as described in paragraph 26.

(6) Before firing, the weapon should be cleaned and oiled as described in paragraph 25.

(7) *The bore and chamber should be entirely free of oil before firing.*

b. Care and cleaning in tropical climates.—(1) In tropical climates where temperature and humidity are high (or where salt air is present) and during rainy seasons, the weapon should be thoroughly inspected daily and kept lightly oiled when not in use. The groups should be dismounted at regular intervals and, if necessary, disassembled sufficiently to enable the drying and oiling of parts.

(2) Care should be taken to see that unexposed parts and surfaces are kept clean and oiled.

(3) Light preservative lubricating oil should be used for lubrication.

c. Care and cleaning in hot, dry climates.—(1) In hot, dry climates where sand and dust are likely to get into the mechanism and bore, the weapons should be wiped clean daily, or oftener if necessary. Groups should be dismounted and disassembled as far as necessary to facilitate thorough cleaning.

(2) When the weapon is being used under sandy conditions, all lubricant should be wiped from the weapon. This

will prevent sand from sticking to the lubricant and forming an abrasive compound which will ruin the mechanism.

(3) Immediately upon leaving sandy terrain, the weapon must be cleaned and relubricated with light preservative lubricating oil.

(4) Perspiration from the hands is a contributing factor to rust because it contains acid. Therefore, metal parts should be frequently wiped dry.

(5) During sand or dust storms, breech and muzzle should be kept covered if possible.

■ 31. CARE DURING GAS ATTACK.—*a. General.*—(1) It is important to prevent the chemicals used in a gas attack from getting in or on the gun and ammunition. When a gas attack is anticipated, steps are taken to cover and protect the gun, ammunition, mount, spare parts, and accessories.

(2) Apply oil to the surfaces of all the parts of the weapon, to the mount, ammunition, and spare parts.

(3) If the gun need not be used during the gas attack, cover the oiled gun with canvas gun covers or place it in a container so that it cannot come into contact with any contaminating chemicals.

(4) After the attack, clean the matériel with dry-cleaning solvent, if uncontaminated.

(5) Prepare for use as described in paragraph 25 or 26, as required.

b. Decontamination.—(1) A complete suit of impermeable clothing and a service gas mask must be worn during decontamination operations.

(2) Matériel contaminated with chemicals other than mustard or lewisite must be cleaned as soon as possible with dry-cleaning solvent or denatured alcohol.

(3) If the surface of the matériel is coated with grease or oil and has been in a mustard or lewisite attack, first remove the grease or oil by wiping with rags wet with dry-cleaning solvent.

(4) Decontaminate unpainted metal surfaces with a solution of *noncorrosive decontaminating agent*. Prepare this by mixing one part of agent to six and one-fourth parts of solvent, or 2 quarts of agent to 3 gallons of solvent. This

agent acts within 15 minutes, and a second application is recommended.

(5) Decontaminate painted surfaces with a bleaching solution. Prepare this by mixing three parts of the bleaching powder (chloride of lime) with two parts of water by volume. The chloride of lime is allowed to remain 24 hours; a second application is made, to insure complete covering, within a brief period after the first application.

(6) After decontamination, clean the matériel thoroughly and prepare for use as described in paragraph 25 or 26, as required.

(7) Do not allow the chemical agents to come into contact with the skin. Always burn or bury all rags or wiping materials used for decontamination.

c. References.—Detailed information on decontamination is contained in FM 21-40 and TM 3-220.

■ **32. POINTS TO BE OBSERVED BEFORE, DURING, AND AFTER FIRING.**—The following list of points to be observed before, during, and after firing will be found useful as a guide for the proper care of the gun. It may also be used as a guide for inspection.

	Before	During	During temporary cessation	After
Bore.....	Look through and clean.....		Clean bore.....	Clean and lightly oil bore with a patch.
Moving parts.....	Oil and test for worn or broken parts. See that parts function without excessive friction.	Keep oiled.....	Inspect and oil; clean dirt from belt holding pawl.	Remove bolt, lock frame, barrel extension, and barrel; clean, oil, and release firing pin.
Head space.....	Make correct adjustment and test (note, par. 11). Examine barrel locking spring.	Tighten if several separated cases occur.	Test.....	Adjust correctly and test.
Rear sight and wind gage.	Clean and free from grease. See that sight is in good mechanical condition. Set sight at 700 and wind gage at zero.	Keep properly set.....	Keep properly set.....	Examine barrel locking spring.
Tripod.....	Set tripod firmly with no lost motion. Verify that traversing bar sleeve latch is properly seated.	Keep firmly set.....	Examine.....	Clean and oil. Set sight at 700 and wind gage at zero.
Belts and ammunition.	Secure sufficient supply of ammunition. Inspect ammunition. Keep belts dry.	Keep belt in line with feed opening. Watch ammunition supply.		Clean, repair, and refill all belts. Separate live rounds from empty cases. Inspect ammunition.
Oil.....	See that oil can is full.....		Refill oil can.....	Refill oil can.
Spare parts and tools.	Keep clean and oiled. See that kits are complete.	Keep within reach.....	Make repairs. Replace broken or worn parts.	Check, replace broken or missing parts, clean, and oil.

SECTION IV

MECHANICAL FUNCTIONING

■ 33. **GENERAL.**—The soldier should have a practical working knowledge of the mechanical operation of the machine gun so that he will be able to keep it in action during combat. Although many parts of the gun operate simultaneously, the subject of functioning is divided into phases to facilitate instruction. The explanation of mechanical functioning begins with the gun assumed to be loaded and ready to fire.

a. *To half load.*—With cover open or closed, enter belt through feedway from left to right. Pull belt through feedway until the first round is in place on the right of belt holding pawl. Close cover if open. Pull bolt fully to the rear and release it.

b. *To load.*—Load is executed the same as half load except that bolt is pulled to the rear and released twice.

c. *To unload.*—Raise cover, remove belt, pull bolt to the rear and hold it, look or feel to see that there is no ammunition in the gun, lower extractor, release bolt, lower cover, and then pull trigger.

d. *To clear gun.*—Raise cover, remove belt, pull bolt to the rear and *hang* it in its rearmost position by engaging extractor cam plunger in rear of extractor feed cam, and inspect the gun to see that there is no ammunition in it.

e. *Division of functioning into phases.*—For a method of dividing functioning into phases and demonstrating each phase, see paragraph 241.

■ 34. **TRIGGER ACTION.**—Since the trigger is pivoted, its forward end is lowered as its rear end is raised. The trigger cams on the front end of trigger, through their engagement with cams on sear, force sear down against the action of sear spring until the shoulder of firing pin is released by sear notch. The firing pin spring then forces firing pin forward to strike primer.

■ 35. **BACKWARD MOVEMENT OF RECOILING PARTS** (figs. 9 and 10).—The explosion forces the recoiling parts (barrel, barrel extension, and bolt) backward about $\frac{5}{8}$ inch. During the

first half of this movement the parts are locked together. When the breech lock clears the breech lock cam, it is forced down by the front projections of the lock frame acting on the breech lock pin. This unlocks the bolt from the barrel extension, permitting the bolt to continue to the rear. As the barrel extension comes to the rear, the barrel plunger spring

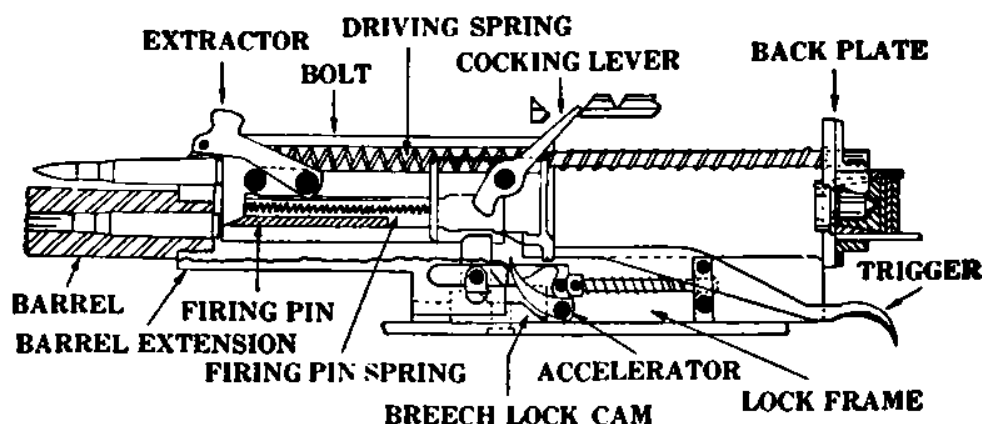


FIGURE 9.—Relation of parts in forward position.

is compressed, and the rear of the barrel extension strikes the accelerator and turns it backward.

■ 36. BACKWARD ACTION OF ACCELERATOR (figs. 9 and 10).—As the accelerator turns backward its tips strike the bottom

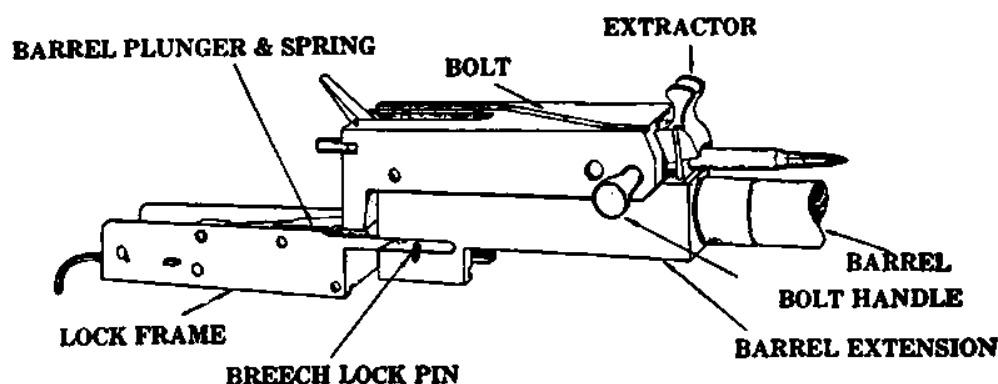


FIGURE 10.—Barrel, bolt, lock frame, and barrel extension in firing position.

projections on the bolt and accelerate it to the rear. The claws of the accelerator engage the shoulder of the T-lug and lock the barrel extension to the lock frame. The barrel plunger spring is thus held compressed. The accelerator stop prevents the accelerator from turning backward too far.

■ 37. **BACKWARD MOVEMENT OF BOLT** (figs. 9 and 11).—As the bolt moves backward the driving spring is compressed. The bolt brings with it a cartridge from the belt, held by the extractor, and an empty case from the chamber, held in the T-slot. The extractor cam plunger rides along the top

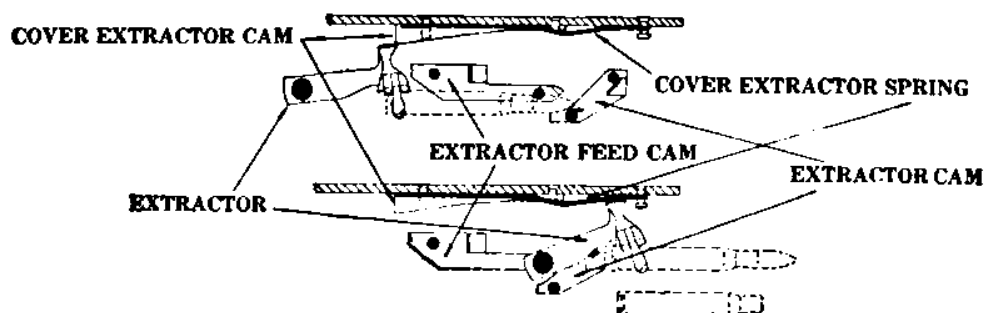


FIGURE 11.—Extracting and loading mechanism.

of the extractor cam and extractor feed cam until forced in by the beveled part of the extractor feed cam. This permits the extractor to be forced down by the cover extractor cam, and the plunger springs out behind the extractor feed cam.

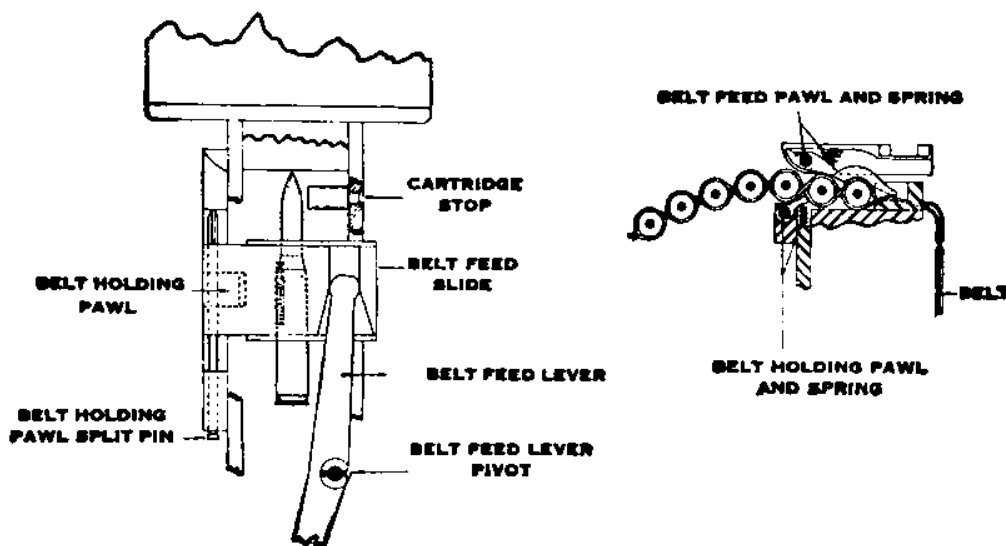


FIGURE 12.—Belt feed mechanism.

■ 38. **FIRST ACT OF FEEDING** (fig. 12).—As the bolt moves backward, the stud on the pivoted belt feed lever moves to the right in the cam groove, thus forcing the belt feed slide to the left. The belt feed pawl engages on the left of the first cartridge which is held in position by the belt holding pawl. In the event the extractor fails to withdraw the lead-

ing round from the belt, the finger of the belt feed pawl, riding on the top of this unextracted round, will hold the feed pawl raised in a position where it cannot engage on the left of the next cartridge. It thus prevents attempted double feeding.

■ 39. COCKING ACTION (fig. 13).—As the bolt moves backward, the upper end of the cocking lever is forced forward in the cocking lever recess, bringing the lower end to the rear. The lower end brings with it the firing pin, thus compressing the firing pin spring against the sear spring pin.

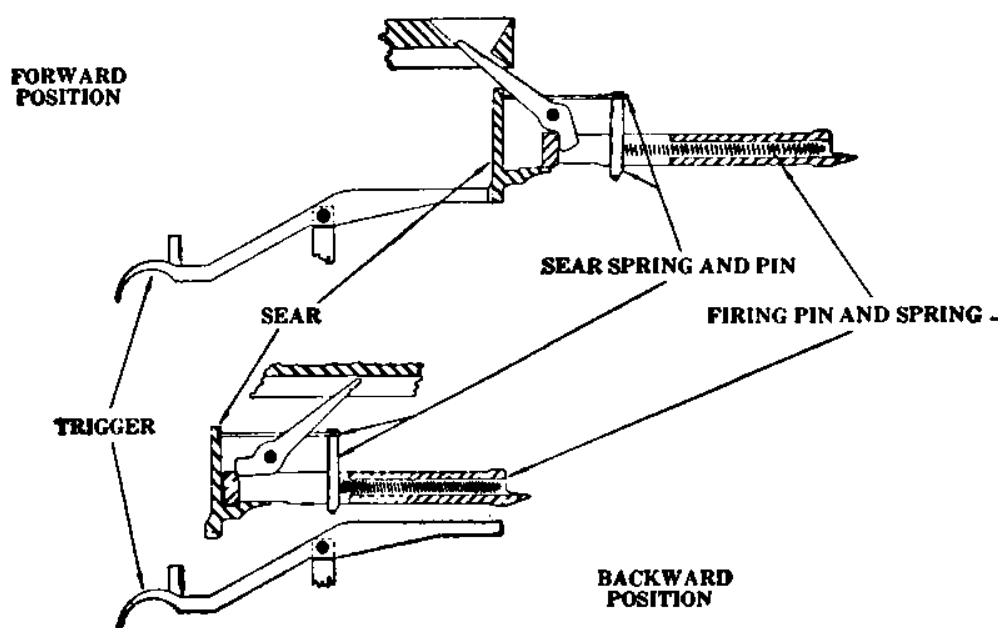


FIGURE 13.—Cocking mechanism.

The shoulder of the firing pin engages the notch in the sear, which is pulled upward by the action of the sear spring, the trigger cams now being disengaged from the sear.

■ 40. ACTION OF DRIVING SPRING (fig. 14).—When the rear of the bolt strikes the buffer plate, its remaining force is absorbed in the coiled spiral spring and the buffer disks. The force of the rearward motion of the bolt being absorbed by the buffer mechanism, the driving spring then forces the bolt forward.

■ 41. FORWARD MOVEMENT OF BOLT (figs. 11 and 13).—When the bolt starts forward, the extractor is guided downward by

the action of the extractor feed cam on the extractor cam plunger. This causes the extractor to force the cartridge down the T-slot in line with the chamber. The ejector knocks the empty case from the T-slot and holds the cartridge in line with the chamber. The upper end of the cocking lever is forced backward, causing the lower end to move forward away from the rear of the firing pin. If the firing pin is prematurely released, it is reengaged by the cocking lever and eased forward so that the striker cannot contact the cartridge primer until after the breech has been locked.

■ 42. RELEASE OF RECOILING PARTS (fig. 14).—As the bolt goes forward its bottom projections strike the accelerator and turn it forward. This unlocks the barrel extension from the lock

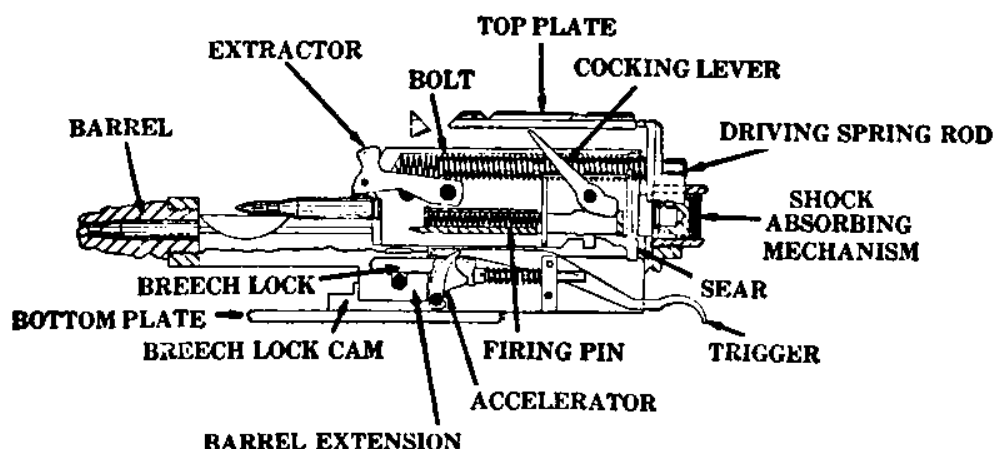


FIGURE 14.—Relation of parts in backward position.

frame and releases the barrel plunger spring. When the accelerator has been tripped, the barrel extension and the barrel are moved forward by the forward force of the bolt acting through the accelerator and by the expansion of the barrel plunger spring.

■ 43. LOADING AND LOCKING ACTION (figs. 11 and 14).—During the latter part of the forward movement of the bolt, the extractor rises as the plunger moves along the top of the extractor cam. When the extractor rises, the ejector is cammed outward, into the half-moon-shaped recess, leaving the cartridge in the chamber gripped by the T-slot. The extractor grips the first round in the belt and is held down firmly by the cover extractor spring, ready to extract the

round. The breech lock is forced upward by the breech lock cam and locks the breech just before the recoiling parts reach the firing position. The breech lock engages in a recess cut in the bottom of the bolt and thus locks the bolt firmly to the barrel extension and against the rear end of the barrel.

■ 44. **SECOND ACT OF FEEDING** (fig. 12).—As the bolt goes forward the stud on the pivoted belt feed lever moves to the left in the cam groove, forcing the slide to the right. The belt feed pawl carries the first cartridge to the right against the cartridge stops, ready to be gripped by the extractor. The next cartridge is carried over the belt holding pawl which rises behind it and holds it in position to be engaged by the belt feed pawl on its next movement to the left.

■ 45. **TRIGGER ACTION IN AUTOMATIC FIRE** (figs. 9, 13, and 14).—If the trigger is held raised, as the sear moves forward with the bolt the trigger cams engage the sear cams and force the sear down, releasing the firing pin. The gun thus fires automatically, repeating the operations of functioning already described. The release of the firing pin actually takes place about $\frac{1}{16}$ inch before the recoiling parts reach the forward position but after the breech is locked.

SECTION V

STOPPAGES AND IMMEDIATE ACTION

■ 46. **DEFINITIONS.**—*a.* A “stoppage” is any unintentional cessation of fire.

b. “Immediate action” is the procedure used for the prompt reduction of usual stoppages.

■ 47. **STOPPAGES.**—*a. Prevention.*—Stoppages will be reduced to the minimum if the gunner has a practical working knowledge of his weapon and applies the points which should be observed before firing. *Prevention is the best remedy for all stoppages.*

b. Causes.—(1) A stoppage will occur if the gun fails to feed, fails to load, fails to fire, or fails to function freely.

(a) If the gun fails to feed, the cause for the stoppage will be found in the ammunition belt or in the feed mechanism.

(b) If the gun feeds but fails to load, the cause will be found in the receiver. A broken part or an obstruction on the T-slot or in the chamber is the usual cause.

(c) If the gun feeds and loads but fails to fire, the cause will be found in the firing mechanism unless the primer of the cartridge is defective.

(2) The table following includes a comprehensive list of possible stoppages. It will serve as a guide during instruction in stoppages or immediate action.

TABLE OF STOPPAGES

Stoppages	Method of preparation for instruction in immediate action and stoppages
1. Misfire due to defective primer.	1. Place a dummy cartridge in belt.
2. Short round.	2. Place a short round in belt.
3. Bulged round.	3. Insert bulged round in belt.
4. Tight loop in belt. ¹	4. Do not prepare.
5. Empty loop in belt.	5. Leave an empty loop in belt.
6. Stretched or torn belt. ¹	6. Do not prepare.
7. Thin rim, permitting nose of bullet to drop below chamber. ¹	7. Do not prepare.
8. Belt improperly loaded.	8. Pull a cartridge partially out of belt.
9. Battered or thick rim of cartridge.	9. Place a battered or a thick-rimmed cartridge in belt.
10. Failure to remove round from chamber.	10. Place a dummy cartridge with rim filed off in the chamber.
11. Set back primer. ¹	11. Do not prepare.
12. Separated case which is removed from chamber by new round when bolt is pulled to the rear.	12. Drive the front portion of a cartridge securely on a dummy cartridge. Pull bolt to the rear and place cartridge properly on the face of bolt. Ease bolt forward.
13. Separated case, which stays in chamber when bolt is pulled to the rear. (Do not set up loose head space.)	13. Insert the front end of a separated case in chamber and load.
14. Bullet loose in cartridge case. Cartridge case extracted from belt but bullet remains in belt. ¹	14. Do not prepare.
15. Short or broken firing pin.	15. Assemble bolt with a defective firing pin or place 5 or 6 successive range dummy cartridges in belt for instruction in immediate action.
16. Weak or broken firing pin spring.	16. Same as 15.

TABLE OF STOPPAGES—Continued

Stoppages	Method of preparation for instruction in immediate action and stoppages
17. Faulty engagement of firing pin and sear notch.	17. Assemble bolt with defective parts.
18. Broken sear spring.	18. Same as 17.
19. Belt feed lever pivot out, worn, or broken.	19. Assemble cover with defective part.
20. Bent or worn belt feed lever.	20. Same as 19.
21. Belt feed pawl spring out or weak.	21. Remove belt feed pawl spring.
22. Belt feed pawl pin out or partially out.	22. Remove belt feed pawl pin.
23. Cover extractor spring out or weak.	23. Remove cover extractor spring.
24. Belt feed lever bent up (stud on lever jumps out of cam groove).	24. Assemble with defective part.
25. Damaged extractor.	25. Same as 24.
26. Belt holding pawl out or spring weak.	26. Remove belt holding pawl.
27. Broken extractor or ejector.	27. Assemble bolt with defective part.
28. Broken or damaged T-slot, causing misalignment and buckling of cartridge as bolt moves forward, or failure to extract. ¹	28. Do not prepare.
29. Weak ejector spring causing misalignment and buckling of cartridge as bolt goes forward.	29. Assemble with defective part.
30. Broken barrel extension or lock frame.	30. Assemble with defective part.
31. Defective trigger mechanism. ²	31. Do not prepare.

¹ Not prepared for instruction in immediate action.

² Defective trigger mechanism may cause the gun to begin firing as soon as it is loaded or to continue to fire when the trigger is released. To remedy, unlatch the cover or twist the belt, unload, disassemble the gun, and replace the defective part or parts.

(3) Sluggish operation of the gun is usually due to excessive friction caused by dirt, lack of proper lubrication, burred parts, tight head space, maladjustment of the buffer assembly, or a weak driving spring.

c. Tools used in reduction of stoppages.—The tools commonly used in the application of immediate action are the ruptured cartridge extractor, the combination tool, and the cleaning rod.

(1) (a) A ruptured cartridge case in the chamber prohibits the entrance of the succeeding round. The ruptured cartridge extractor, commonly called "clearing plug," is used to

remove the ruptured case. This tool is furnished in two types, the cartridge form and the Mk. IV.

1. To use the cartridge form, open cover, place ruptured cartridge extractor on feedway against cartridge stops, and engage it with extractor on bolt. Holding down on extractor, load it *slowly* and *smoothly* into chamber, being careful to see that the shoulder does not strike and bur the edge of chamber. Strike forward on bolt handle to be sure that ruptured cartridge extractor is seated in ruptured case. Pull bolt to the rear and catch ruptured cartridge extractor in the left hand as it is ejected.
2. To use the Mk. IV, draw bolt handle to the rear and hang it. Insert nose of ruptured cartridge extractor in chamber, handle up, push it forward to seat it firmly in the separated case, then pull backward on handle.

(b) Separated cases should be removed from the ruptured cartridge extractor so that it will again be ready for use. To do this, the end of the ruptured cartridge extractor must be unscrewed, the separated case taken off, and the end screwed on again.

(2) The use of the combination tool facilitates the removal of a round which is stuck on the T-slot. To remove a round from the T-slot, pull bolt fully to the rear and raise extractor. If the round does not fall out, remove it with the combination tool. If the round is above the bolt handle slot, insert screw driver end of tool through slot and under the rear end of cartridge and pry it up. If the round is below the slot, place screw driver end of tool through top of receiver into cannellure of cartridge and drive it down.

(3) The cleaning rod is used to remove a case from the chamber which has not been extracted by the bolt. To remove the case from the chamber, pull bolt fully to the rear and hold it. Remove the following round from the T-slot. Grasp cleaning rod about 6 inches from the end, insert it into muzzle, and force case from chamber. Care should be used not to damage the face of the bolt by striking it as the rod is shoved through the bore.

d. Inspection of feed mechanism.—In event of recurrent or permanent failure to feed, the feed mechanism must be inspected and the defective part replaced. This inspection should include the following:

(1) *Belt feed lever and slide.*—(a) With the bolt forward, lower cover, making sure that stud on feed lever is in cam groove in the top of bolt.

(b) Press to the left on belt feed slide to remove the play and note whether or not the slide protrudes to the right of the cover about $\frac{1}{32}$ inch.

1. If it does not, change belt feed lever and pivot.

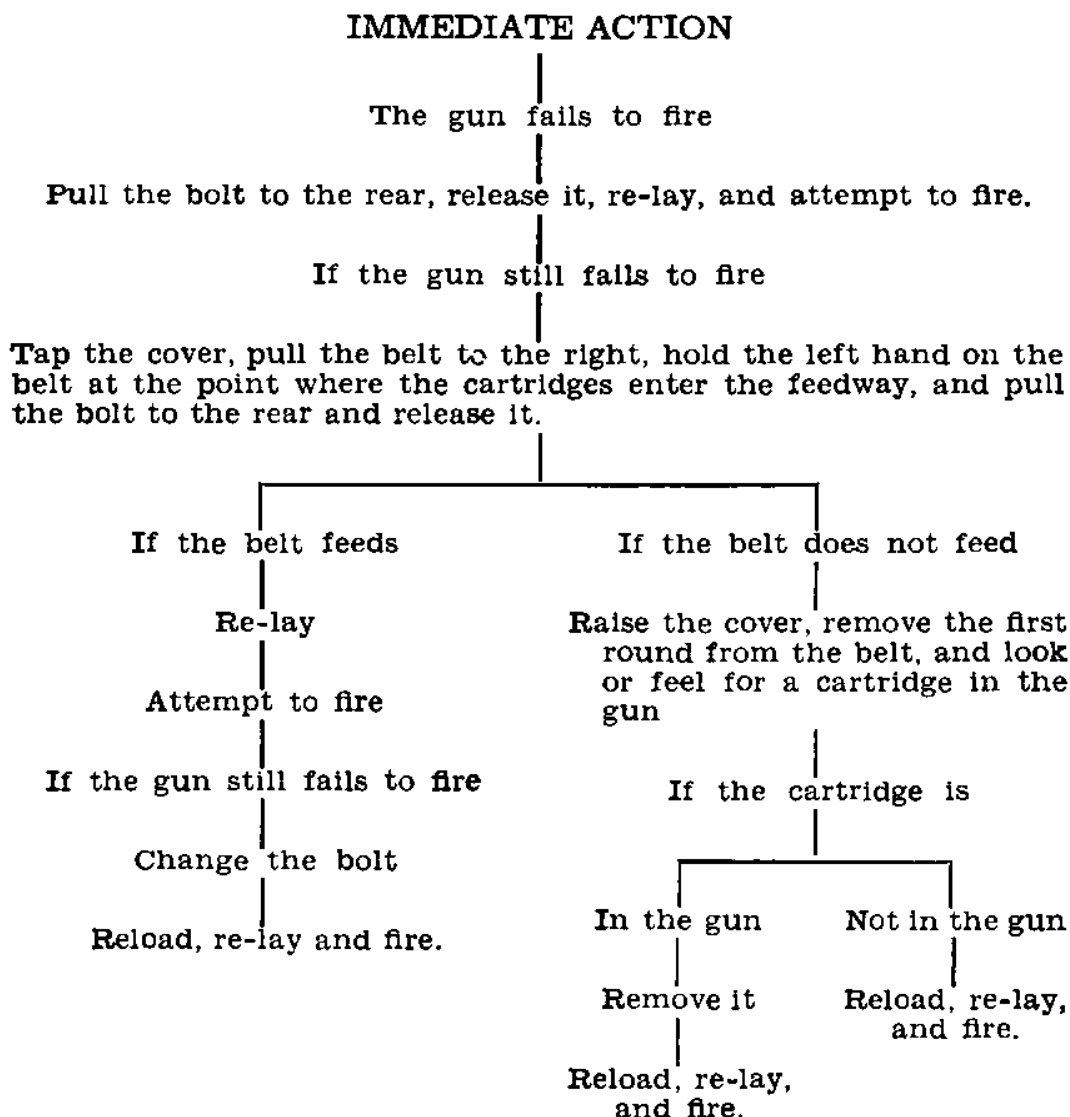
2. If it does, and the gun has occasionally failed to feed, examine belt feed lever for excessive wear at the pivot and stud. Also see that stud projects a sufficient distance from cover to insure that it is positively positioned in cam groove in the top of bolt when cover is closed. Replace defective parts.

NOTE.—If the belt feed slide does not extend $\frac{1}{32}$ inch beyond the right side of the cover, the cartridge may not be correctly positioned and the extractor may strike the base of the cartridge, force the bullet into the case, and cause a short round. The resulting failure to load the cartridge is thus caused by a worn or bent belt feed lever, or a defective slide. The base of such a cartridge will show evidence of having been hit by the extractor.

(2) *Other parts.*—If the slide positions properly and no excessive play exists, examine cover extractor spring; belt feed pawl, spring, and pin; belt holding pawl and spring; extractor including ejector; spring and cam plunger; and T-slot, to see that these parts are there, in good condition, and that springs are sufficiently strong to actuate the parts properly.

■ **48. IMMEDIATE ACTION.**—*a. General.*—The procedure prescribed in immediate action for the reduction of stoppages is based on the frequency with which the various types of stoppages occur. Execution of this procedure by the gunner will enable him to remedy the majority of stoppages immediately without attempting to analyze the cause. Immediate action is performed by the gunner. All personnel required to fire the machine gun should be proficient in immediate action. Immediate action must be carried out quickly and, with practice, should become instinctive.

b. Procedure.—The procedure for immediate action is shown in the following diagram:



NOTES.—1. If the procedure outlined above does not place the gun in action, the gunner must inspect the feed mechanism and the working parts of the gun to include inspecting the chamber for a ruptured case in order to locate and reduce the stoppage.

2. If the bolt stops just short of *home*, frequently it can be pushed fully forward and firing continued without application of immediate action. *Caution:* Release the trigger before pushing the bolt forward.

3. Stoppages caused by defective feed mechanism can sometimes be remedied temporarily by pulling the belt to the right with just sufficient force to assist the feed mechanism in positioning the leading round.

4. Immediate action is not complete until the gun has been re-layed and fired.

SECTION VI

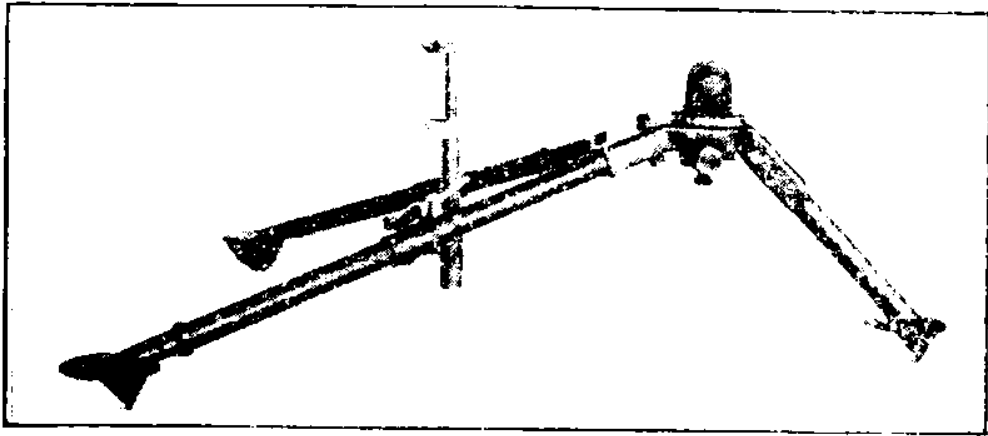
TRIPOD MOUNTING

■ 49. LIGHT MACHINE GUN TRIPOD MOUNT M2.—*a.* This tripod (fig. 15) is a simple mount for the light machine gun and is designed to furnish the gun with a stable mount embodying the maximum strength and rigidity consistent with desired light weight. It is portable, easily packed, and may be used for the delivery of accurate fire on ground targets.

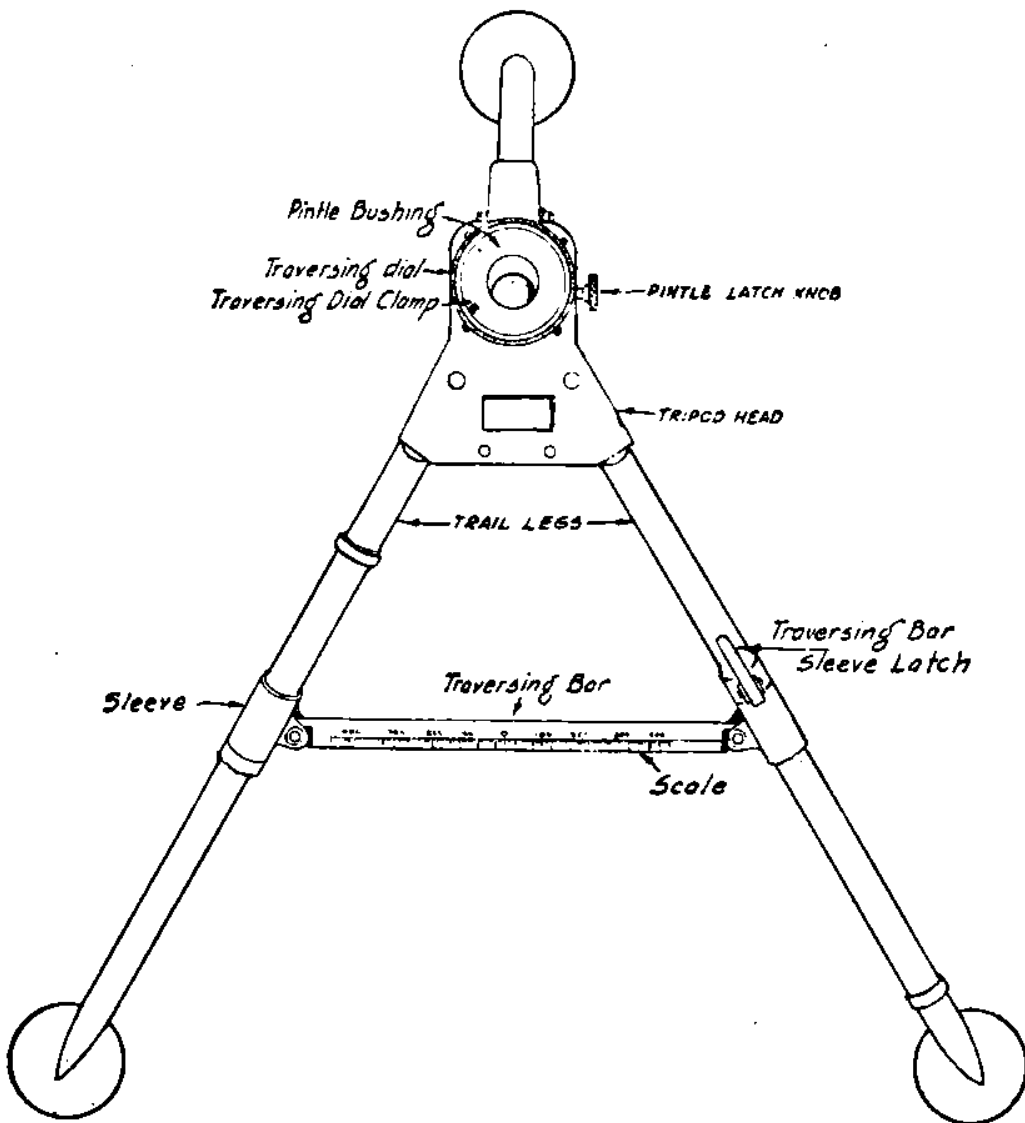
b. The tripod consists of three tubular steel legs articulating in a tripod head, the two trail legs being joined and additionally supported by a traversing bar, forming a simple A-truss, and serving as a rear support for the mounted gun. The tripod head furnishes a front support for the mounted gun, it being in turn supported by the short front leg.

c. Incorporated in the tripod head (fig. 16) is a bronze bushing, mating with the tapered steel pintle permanently secured to the receiver of the light machine gun. The pintle is secured in its union with the bushing by engagement of the pintle latch of the tripod in its corresponding annular groove of the pintle (upper groove). The pintle latch is spring actuated and is seated in its housing on the lower right surface of the tripod head.

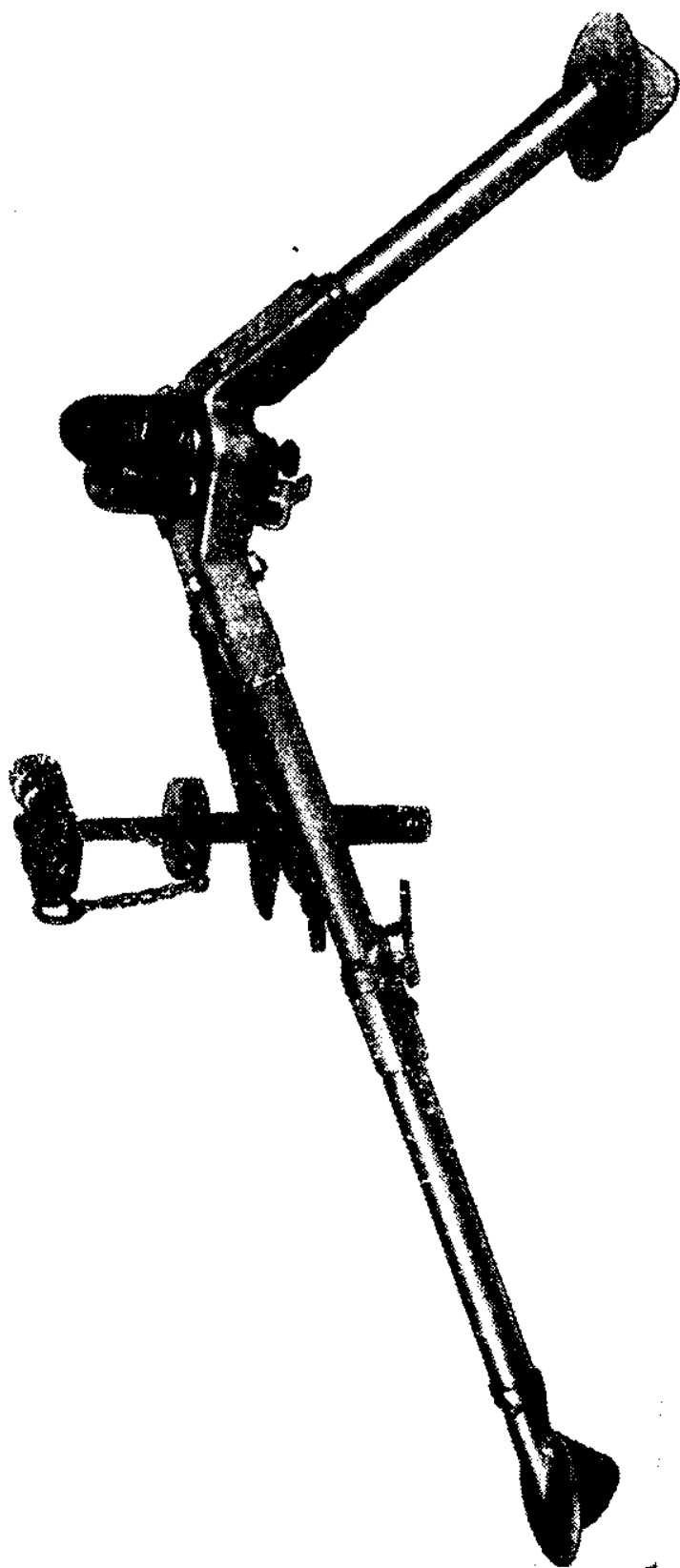
d. Tripods manufactured prior to the summer of 1942 have a traversing dial graduated in major divisions of 100 mils and minor subdivisions of 25 mils, from zero to 3,200 mils in both directions, fitted to the head of the tripod by the flange of the bronze pintle bushing. This dial is free to rotate but may be clamped in any desired position by the traversing dial clamp. This clamp is actuated by the knurled head of the traversing dial clamp which is located under the left lower surface of the tripod head. A zero index engraved on the left rear quadrant of the pintle is utilized in conjunction with the traversing dial to establish or measure horizontal angles.



① Early type machine-gun tripod M2.

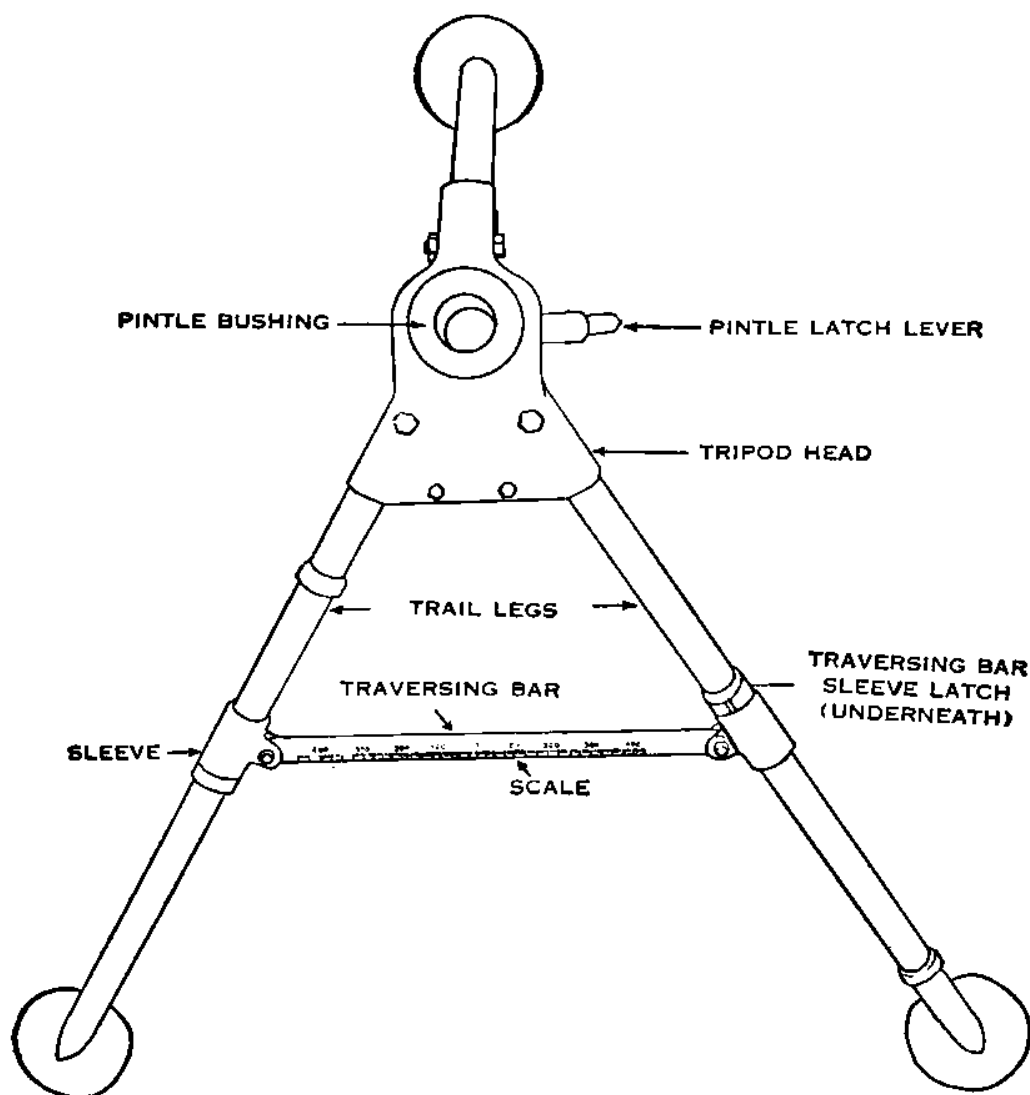


② Early type machine-gun tripod M2—schematic.
FIGURE 15.



① Late type, machine-gun tripod M2.
FIGURE 16.

e. The traversing bar provides a rear support for the mounted gun, the traversing slide of the elevating and traversing mechanism mating with the bar. An additional device for measuring or establishing horizontal angles is incorporated in the scale engraved on the traversing bar. This scale is divided into 100-mil major divisions and 5-mil



② Late type, machine-gun tripod M2—schematic.

FIGURE 16—Continued.

minor subdivisions. It has a range of 450 mils to the right and 425 left of the zero graduation.

f. The traversing bar articulates in sliding sleeve devices at each end, the sleeves moving along the trail legs in mounting and dismounting the tripod. Positioning stops are incorpo-

rated on the trail legs, positively positioning the traversing bar in both the mounted and dismounted positions of the tripod. The traversing bar sleeve latch, mounted on the right trail leg, secures the traversing bar in its proper position when the tripod is mounted.

■ 50. **DISASSEMBLING AND ASSEMBLING OF LIGHT MACHINE GUN TRIPOD M2.**—*a.* The disassembly and assembly of the parts of the light machine gun tripod are not essential except to replace broken or damaged parts.

b. The front leg articulation with the tripod head, the rear leg articulations with the tripod head, and the articulations of the traversing bar with its sleeves are standard commercial bolts with lock washers and nuts to fit. Spare bolts, nuts, and washers are carried in the spare parts chest. All these unions must be kept tight to insure accurate firing.

c. The pintle latch housing is secured to the tripod head by standard commercial cap screws and suitable lock washers. Spare cap screw and washer and pintle latch spring are carried in spare parts chest.

d. The traversing bar-sleeve latch is pivoted on the right trail leg by rivet assembly.

e. The pintle bushing is of bronze and is a pressed fit into the tripod head. It may be removed by pressing it out and may be replaced by new bushing. Removal of the pintle bushing frees the traversing dial. This disassembly is preferably done by Ordnance Department personnel.

f. The traversing dial clamping mechanism, seated in its recess in the tripod head, may be removed by first removing the small hexagonal lock nut on the foot of the clamp, then the knurled clamp nut. This permits the clamp being removed from the top of the tripod head. This disassembly is preferably done by Ordnance Department personnel.

SECTION VII

ACCESSORIES

■ 51. **PACK EQUIPMENT.**—Special pack equipment for use with the light machine gun consists of—

- 1 tripod and gun hanger

- 1 ammunition hanger
- 3 ammunition chests
- 1 spare parts chest
- 1 gun cover, canvas
- 1 spare barrel canvas case.

■ 52. HANGERS.—*a. Tripod and gun hanger.*—(1) This pack hanger is a light duralumin and steel hanger, accommodating the tripod, the light machine gun, and the spare parts chest. It fits the standard Phillips cavalry pack saddle and is carried on the right side, hung on the horizontal supporting member of the saddle by steel reinforced hooks of the main supporting members of the hanger and secured by two hold-down straps.

(2) The spare barrel canvas case with contents is habitually carried strapped to the lower horizontal member of the hanger to balance the pack load laterally. The case should be carried with the opening to the rear. When ammunition has been expended, the case with contents may be strapped to the lower horizontal member of the ammunition hanger on the left side to retain lateral balance of the pack load.

b. Ammunition hanger.—This hanger is of duralumin and steel construction, accommodating three ammunition chests, and is habitually carried on the left side of the standard Phillips cavalry pack saddle. The hanger is symmetrical longitudinally and may be used on either side of the Phillips cavalry pack saddle in case an extra ammunition pack horse is utilized to transport ammunition for the light machine guns of a horse cavalry organization.

■ 53. CHESTS.—*a. Ammunition chests.*—The ground light machine gun is provided with ammunition chest M1, or the same chest adopted for use with the machine gun M1917A1.

b. Spare parts chest.—The authorized contents of the spare parts chest are listed in SNL No. A-6.

■ 54. COVERS.—*a.* The canvas gun cover is provided to protect the gun.

b. The spare barrel canvas cover is designed to accommodate an assembled barrel and barrel extension assembly.

■ 55. BELT FILLING MACHINE (fig. 17).—*a. General.*—This machine is for the rapid filling of fabric (web) belts. It is normally fastened to a table or bench while in use.

b. To fill a web ammunition belt with cartridges.—(1)

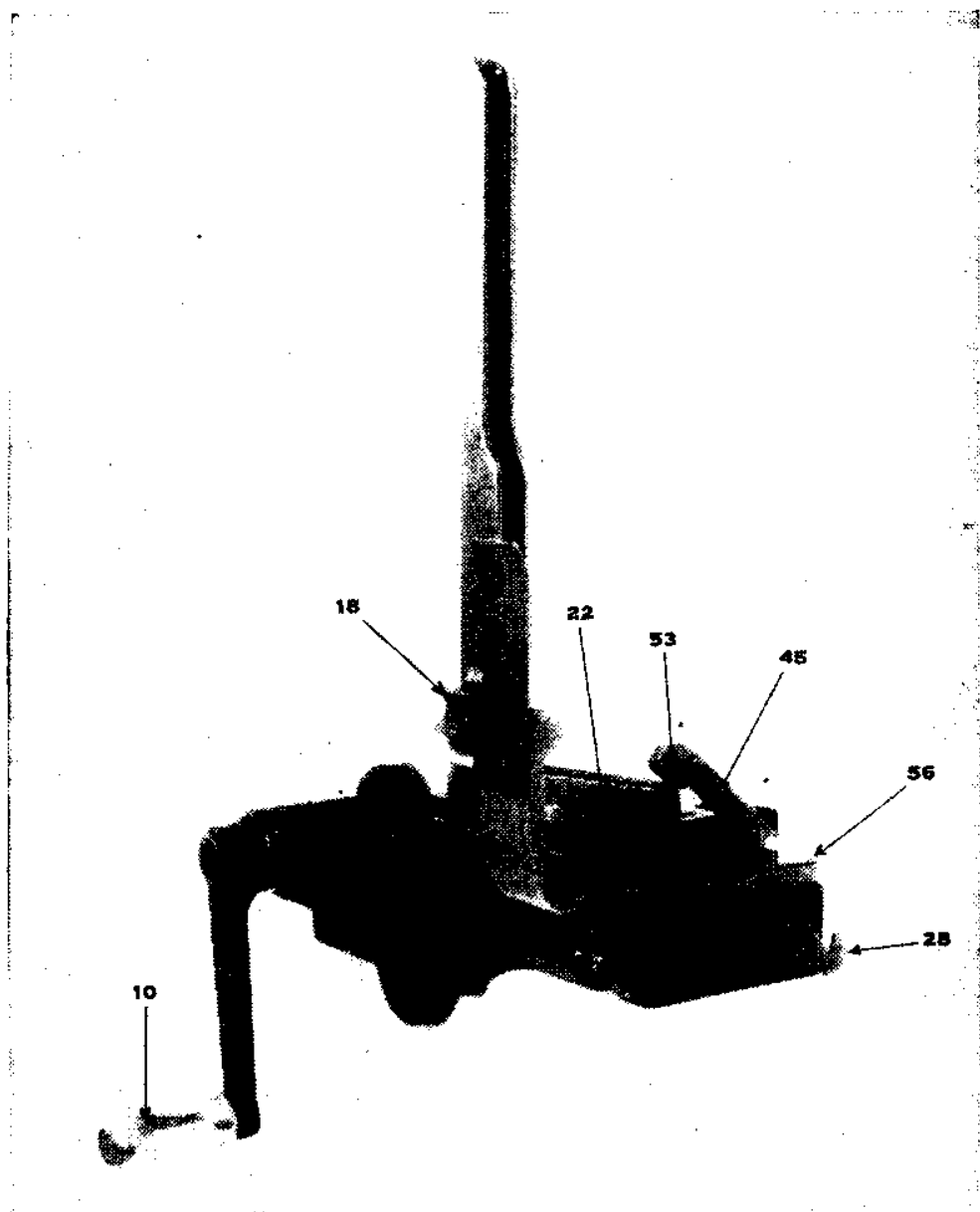


FIGURE 17.—Belt filling machine (web belts).

Fasten machine to a table or bench and turn crank (10) to the *right* until it is straight down.

(2) Release tension spring hook (28) and raise upper feed wheel (18) as far as it will go.

(3) Turn belt guide cover (56) to the right far enough to

admit belt into belt guide (53) and raise upper needle bar (45) as far as it will go.

(4) A black thread running along the side of the belt indicates the side for the entrance of the cartridge. Put two cartridges by hand into the two loops of the belt nearest the end with the brass tip, and place belt in machine with first cartridge resting in top groove of lower feed wheel (22), belt passing out at back through belt guide.

(5) Return belt guide cover to place over belt (being careful to see that the belt is free to pass under it), and lower needle bar.

(6) Turn upper feed wheel down upon belt and secure tension spring under hook.

(7) Fill feed guide with cartridges by stripping ten at a time from the paper boxes in which they are packed.

(8) Turn crank to the right and the cartridges will be fed into the belt ready for use in the gun.

(9) Place a feed box ready to receive the filled belt and at such a height that not more than 2 feet of filled belt will be suspended from the feed wheels of the machine.

c. Precautions for using the belt filling machine for web belts.—(1) Before using the machine, see that it is well oiled, that all screws are tight, and that the needles are properly set. See that the points of the upper needles are about 0.01 inch vertically above the points of the lower needles.

(2) In case of a miss in loading the belt, stop and open up machine and remove belt. Turn crank to the right until straight down, as in starting, and replace belt in machine with the *next to the last* cartridge in top groove of lower feed wheel. Close machine and proceed as before.

(3) After use, the machine should be cleaned, oiled, adjusted, and replaced in its wooden box.

■ 56. BLANK AMMUNITION ATTACHMENTS (fig. 18).—*a. General.*—The blank ammunition attachments for use with the machine gun have been developed in order to simulate fire. The attachments are designed to fire blank ammunition M1909.

(1) The *cartridge stop attachment* is inserted by withdrawing the belt holding pawl split pin about $\frac{1}{2}$ inch and sliding the attachment over the cartridge stop, lining up the small hole with the belt holding pawl split pin and replacing this pin. The cartridge stop attachment acts as a guide for the blank cartridge and also as a cartridge stop. Its chief use is to hold the cartridge in position in feedway so that the extractor may engage its cannellure; it also prevents the entrance of a live round into the feedway.

(2) The *muzzle attachment* is assembled to the gun by removing the front barrel bearing, sliding the threaded end of the attachment over the barrel, and screwing it into the barrel jacket.

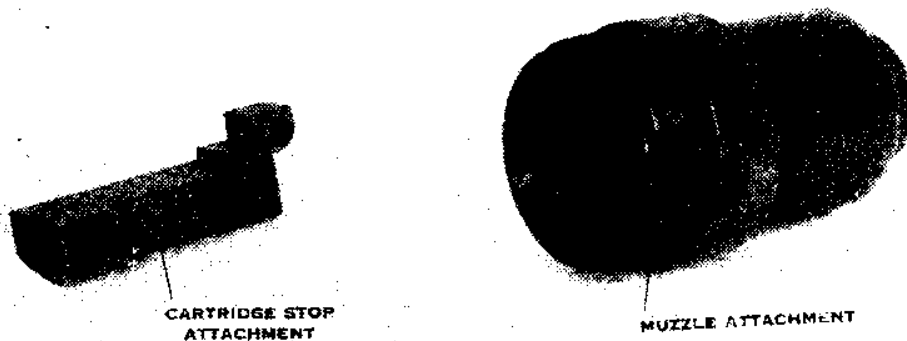


FIGURE 18.—Blank firing attachments.

b. Precautions.—(1) Never attempt to use the muzzle attachment unless the cartridge stop attachment is in its proper place in the feedway.

(2) Always remove the muzzle attachment before removing the cartridge stop attachment in order to eliminate the possibility of firing ball ammunition with the muzzle attachment still in position.

(3) See that the muzzle attachment is clean inside before using.

(4) After firing blank ammunition, clean the barrel and muzzle attachment. It is absolutely essential that this be done before firing any other type of ammunition.

■ 57. SPARE BARREL.—A spare barrel is issued for each light machine gun. In pack, this barrel is carried on the gun and tripod hanger. Under field conditions the barrel, barrel extension assembly, barrel locking spring, breech lock, and breech lock pin assembly should be assembled into one unit and carried in the spare barrel cover.

■ 58. ASBESTOS MITTENS.—One pair of asbestos mittens is issued for use with each machine gun to facilitate the handling of the gun when hot, particularly when moving to a new position by hand, and in changing barrels.

SECTION VIII

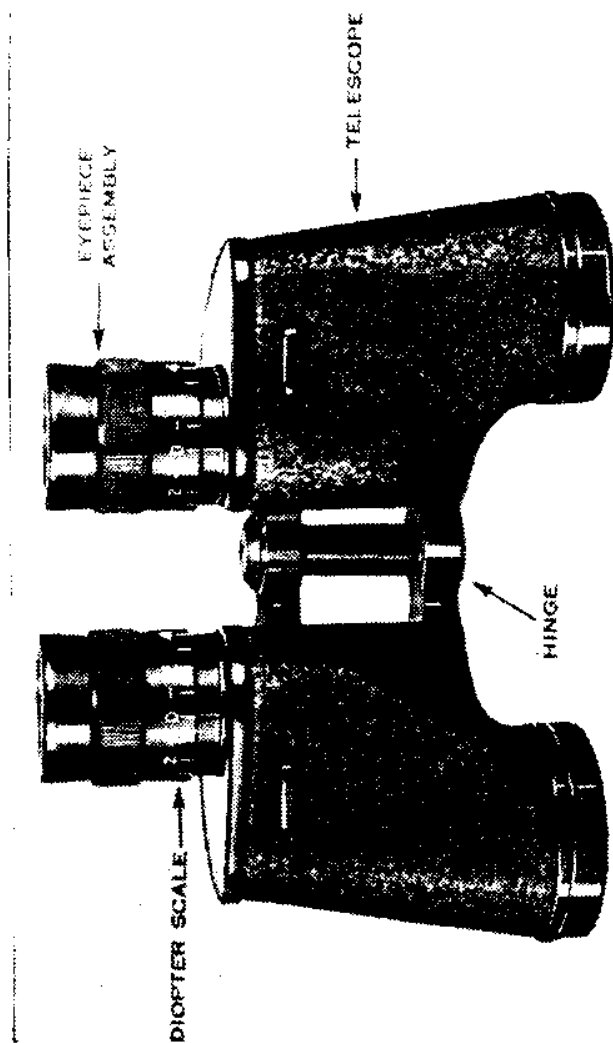
FIRE CONTROL INSTRUMENTS

■ 59. BINOCULARS.—There are two binoculars in use in the service: the M3 (fig. 19 ①) and the type EE (fig. 19 ☺). The M3 will eventually replace the type EE. The paragraphs below apply to both models. Each model consists of the binocular with its carrying case.

a. Description.—(1) The binocular consists of two compact prismatic telescopes pivoted about a common hinge which permits adjustment for interpupillary distances. A scale, graduated every 2 millimeters from 56 to 74, permits the observer to set the telescope rapidly to suit his eye distance when the spacing of his eyes is known. The eyepieces can be focused independently for each eye by screwing them in or out. Each eyepiece is provided with a diopter scale for rapid setting when the observer knows the correction for his eye. The zero graduations indicate the settings for normal eyes.

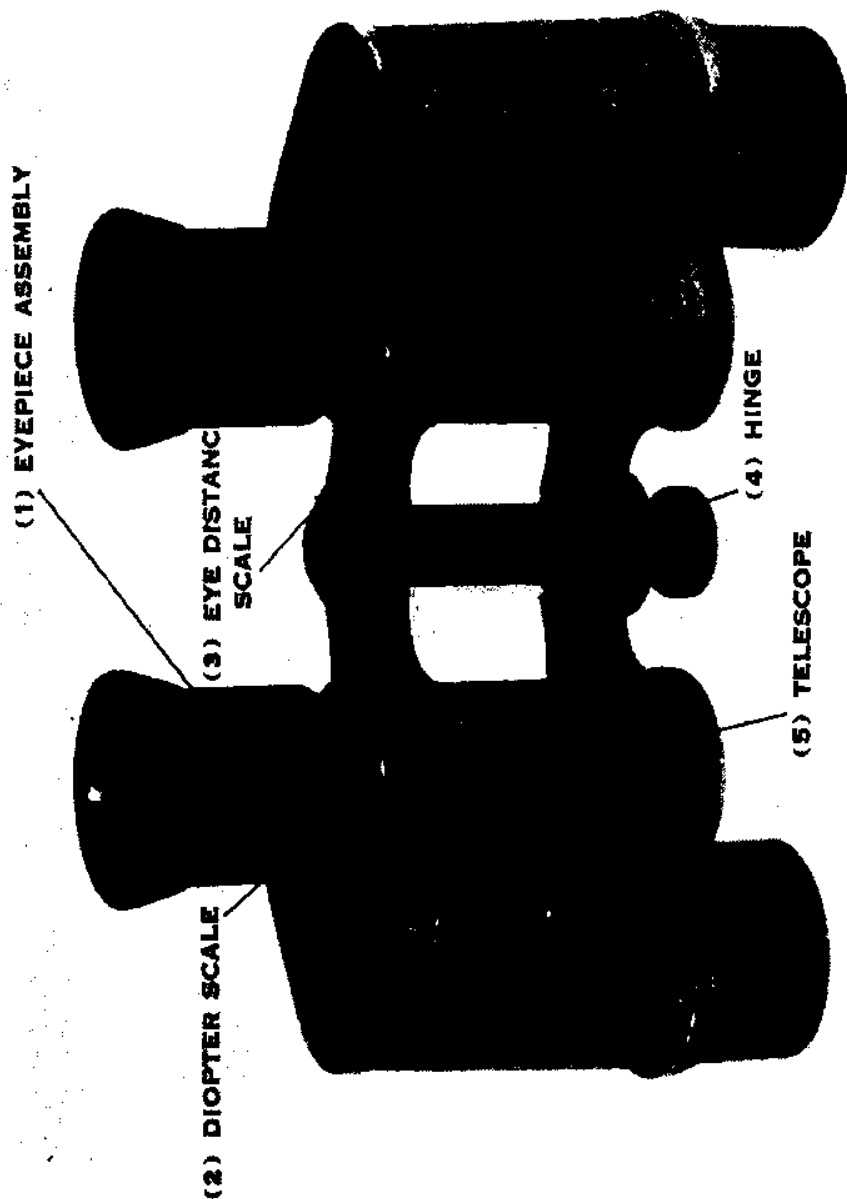
(2) The left telescope is fitted with a glass reticle (fig. 20) upon which are etched a vertical mil scale, a horizontal mil scale, and a stadia graduated similarly to the sight leaf graduation on the service rifle, but inverted.

b. Use.—The binocular is used for observations and the measurement of small horizontal and vertical angles in mils. The vertical stadia scale is used to pick up auxiliary aiming marks in direct laying and to determine troop safety for overhead fire.



① Binocular M3.

FIGURE 19.—Fire control instruments.



③ Binocular type EE.

FIGURE 19.—Fire control instruments—Continued.

c. Preliminary adjustments.—(1) *Interpupillary distances.*—To adjust the binocular so that the eyepieces are the same distance apart as the pupils of the observer's eyes, point the binocular at the sky and open or close the hinged joint until the field of view ceases to be two overlapping circles and appears as one sharply defined circle. Note the reading on the scale which indicates the spacing of the eyes. This setting will be the same on any other M3 or type EE binocular for the same individual.

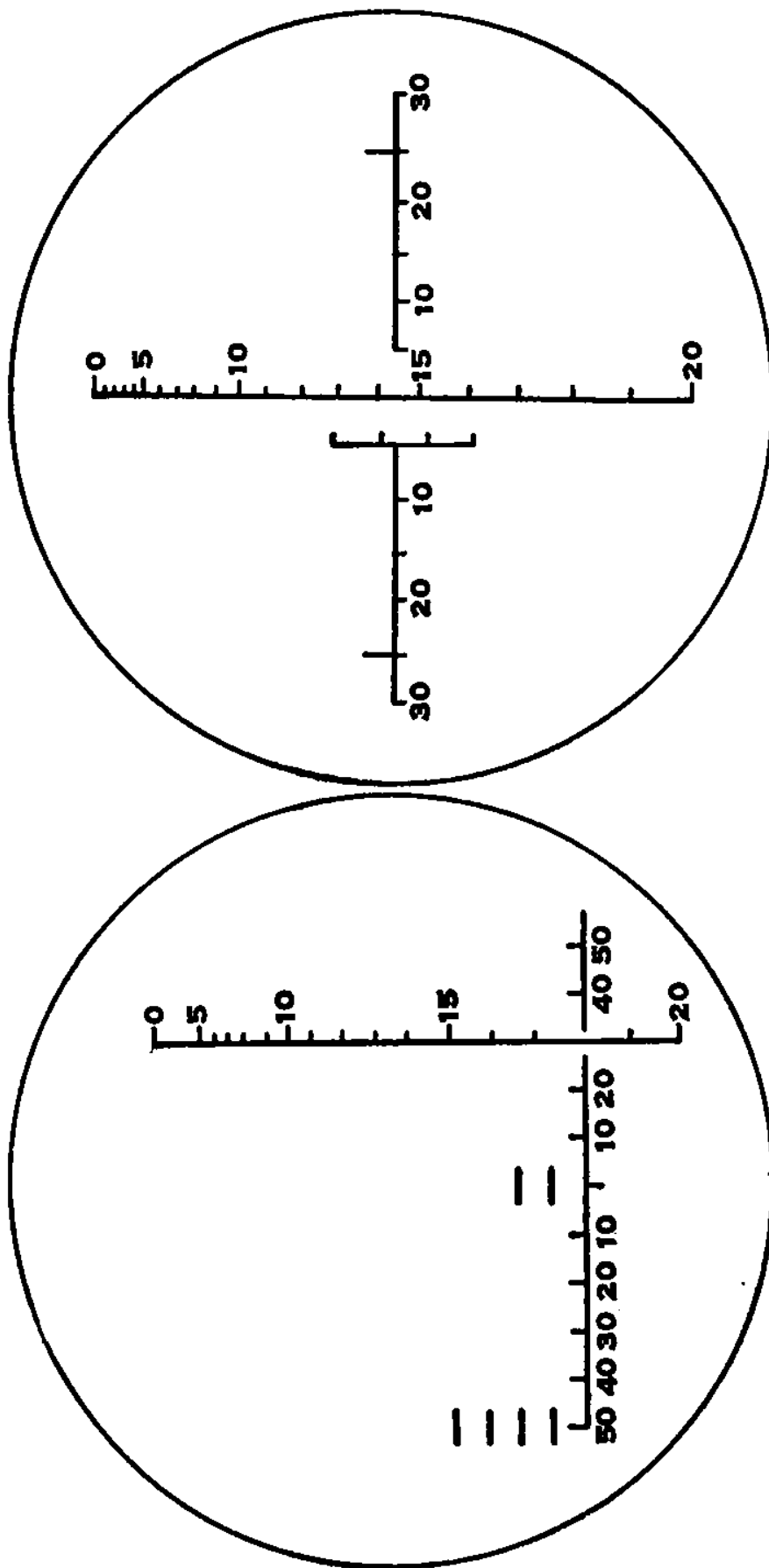
(2) *Focus of eyepiece.*—Look through the binocular, with both eyes open, at an object several hundred yards away. Place the hand over the front of one telescope and screw the eyepiece of the other in or out until the object is sharply defined. Repeat this operation for the other eye, and note the reading on each diopter scale. This setting will be the same on any other M3 or type EE binocular for a given individual.

d. Operation.—(1) In using the binocular, it should be held in both hands and pressed lightly to the eyes so as to keep the relation with the eyes constant without transmitting tremors from the body. The bent thumb should fit into the outer edges of the eye sockets in such a manner as to prevent light from entering in rear of the eyepieces. When possible it is best to use a rest for the binocular or for the elbows.

(2) The mil scales are seen when looking through the binocular end, by superimposing them upon the selected objects, the horizontal and vertical angles between these objects may be read.

(3) The inverted sight leaf scale of the type EE binocular is used to secure range settings on sharply defined auxiliary aiming points when the target is not clear enough for direct aiming. This procedure is described in paragraph 199.

e. Care.—The binocular is a rugged, serviceable instrument but it should not be abused or roughly handled. Do not wipe the lenses with the fingers. Use only clean lens tissue paper to wipe the lenses if they become dirty. No substitute should ever be used for this purpose. The use of polishing liquids or pastes for polishing lenses is forbidden.



② Binocular type EE.

① Binocular M3.

FIGURE 20.—Reticles.

■ 60. **WRIST WATCH.**—The wrist watch is a timepiece of the usual construction supplied with a strap for fastening to the wrist.

SECTION IX

INDIVIDUAL SAFETY PRECAUTIONS

■ 61. **PRECAUTIONS FOR DISASSEMBLING AND ASSEMBLING GUN.**—Paragraphs 12 and 13 outline the procedure for disassembly and assembly of the driving spring rod to the bolt. In order to avoid serious injury to personnel, the precautions outlined should be carefully observed.

■ 62. **BLANK FIRING ATTACHMENT.**—*a.* Never attempt to use the muzzle attachment unless the cartridge stop attachment is in its proper place in the feedway.

b. Always remove the muzzle attachment before removing the cartridge stop attachment in order to eliminate the possibility of firing ball ammunition with the muzzle attachment still in position.

c. See that the muzzle attachment is cleaned inside before using.

■ 63. **HOT GUN.**—*a.* To avoid serious burns, care must be exercised that the barrel and jacket are not touched with bare hands. For moving the gun or disassembling a hot barrel, the asbestos mittens should be used.

b. If the barrel becomes overheated after protracted periods of firing, the gun should be cleared immediately on suspending or ceasing fire. The heat of the barrel will cause the gun to fire without pulling the trigger, if a round is left a short time in the chamber of an overheated barrel.

■ 64. **SAFETY PRECAUTIONS ON THE RANGE.**—Safety precautions for range firing are included elsewhere in this manual (pars. 130, 173, 252*e*) and are not repeated here.

SECTION X

AMMUNITION

■ 65. **GENERAL.**—The information in this section pertaining to the several types of cartridges authorized for use in the Browning machine gun, caliber .30, M1919A4, includes a de-

scription of the cartridges, means of identification, care, use, and ballistic data.

■ 66. CLASSIFICATION.—*a.* Based upon use, the principal classification of the ammunition for this machine gun is—

(1) Ball, for use against personnel and light matériel targets.

(2) Tracer, for observation of fire and incendiary purposes.

(3) Armor-piercing, for use against armored vehicles, concrete shelters, and similar bullet-resisting targets.

b. Other types provided for special purposes are—

(1) Blank, for simulated fire.

(2) Dummy, for training (cartridges are inert).

■ 67. LOT NUMBER.—When ammunition is manufactured, an ammunition lot number which becomes an essential part of the marking is assigned in accordance with pertinent specifications. This lot number is marked on all packing containers and on the identification card inclosed in each packing box. It is required for all purposes of record, including grading and use, and reports on condition, functioning, and accidents in which the ammunition might be involved. Only those lots of grades appropriate for the weapon will be fired. Since it is impractical to mark the ammunition lot number on each individual cartridge, every effort will be made to maintain the ammunition lot number with the cartridges once they are removed from their original packing. Cartridges which have been removed from the original packing and for which the ammunition lot number has been lost are placed in grade 3. It is therefore obvious that when cartridges are removed from their original packings they should be so marked that the ammunition lot number is preserved.

■ 68. GRADE.—Ordnance Field Service Bulletin No. 3-5 explains the meaning of grading for caliber .30 ammunition. This bulletin also lists numerically every lot of ammunition with its correct grade as established by the office of the Chief of Ordnance. Only lots of proper grade will be fired. Grade 3 ammunition may be fired only under special provisions of Ordnance Department Circular No. 205, 1942.

■ 69. IDENTIFICATION.—*a. Markings.*—The contents of original boxes are readily identified by the markings on the box. Similar markings on the carton label identify the contents of each carton.

b. Color bands.—Color bands painted on the sides and ends of the packing boxes further identify the various types of ammunition. The following color bands are used:

Cartridge, ball.....	Red.
Cartridge, ball, and cartridge, Composite band of yellow, tracer, in metallic link belts. ¹	red, and green stripes (yellow on left, red in center, green on right).
Cartridge, tracer.....	Green on yellow.
Cartridge, armor-piercing.....	Blue on yellow.
Cartridge, blank.....	Blue.
Cartridge, dummy.....	Green.

¹ Special packing for Army Air Forces, not for use in the Browning machine gun, caliber .30, M1919A4.

c. Types and models of caliber .30 cartridges.—When removed from their original packing containers, the cartridges may be identified, except as to ammunition lot number and grade, by physical characteristics described below:

(1) *Ball.*—All models of caliber .30 ball ammunition, except M2 of 1940 and earlier, and M1906, have bullets with gilding metal packets. The jacket of the M1906 bullet is cupronickel which has a silvery appearance. The gilding metal jacket of the early M2 ammunition is tin coated and hence resembles the M1906 bullet in appearance. The gilding metal jacket of the present M2 bullet is copper-colored.

(2) *Tracer.*—Caliber .30 tracer ammunition may be identified by the nose of the bullet, which is painted red for a distance of approximately $\frac{1}{4}$ inch from the tip.

(3) *Armor-piercing.*—All models of caliber .30 armor-piercing ammunition may be distinguished by the nose of the bullet, which is painted black for a distance of approximately $\frac{1}{4}$ inch from the tip. The bullets have gilding metal jackets.

(4) *Blank.*—Blank ammunition may be identified by the absence of a bullet in the cartridge case.

(5) *Dummy.*—The caliber .30 corrugated dummy cartridge

may be identified by six longitudinal corrugations and a hole in the cartridge case.

■ 70. CARE, HANDLING, AND PRESERVATION.—*a.* Small arms ammunition, as compared with other types, is not dangerous to handle. Care, however, must be observed to keep the boxes from becoming broken or damaged. All broken boxes must be immediately repaired and careful attention given so that all markings are transferred to the new parts of the box. The metal liner should be air-tested and sealed if equipment for this work is available.

b. Ammunition boxes should not be opened until the ammunition is required for use. Ammunition removed from the airtight container, particularly in damp climates, is likely to corrode, thereby becoming unserviceable.

c. Protect the ammunition carefully from mud, sand, dirt, and water. If it gets wet or dirty, wipe it off at once. Verdigris or light corrosion, if it forms on cartridges, should be wiped off. However, cartridges should not be polished to make them look better or brighter.

d. The use of oil or grease on cartridges is dangerous and is prohibited.

e. Do not fire dented cartridges, cartridges with loose bullets, or otherwise defective rounds.

f. Do not allow the ammunition to be exposed to the direct rays of the sun for any length of time. This is likely to affect its firing qualities seriously.

■ 71. STORAGE.—*a.* Whenever practicable, small arms ammunition should be stored under cover. This applies particularly to tracer ammunition which is subject to rapid deterioration if it becomes damp. It may even ignite spontaneously. Should it become necessary to leave small-arms ammunition in the open, it should be raised at least 6 inches from the ground, and the pile covered with a tarpaulin. Suitable trenches should be dug to prevent water flowing under the pile.

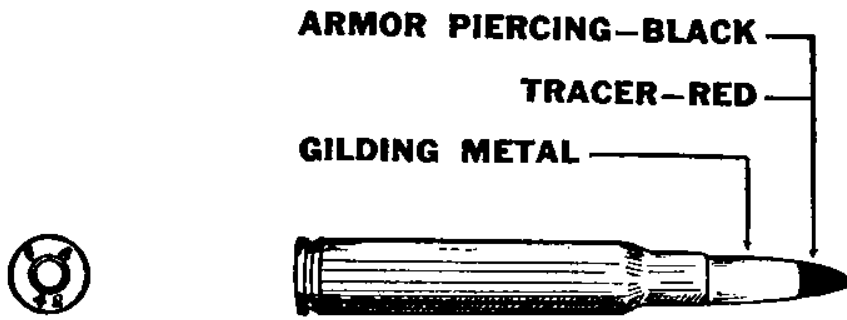
b. If practicable, tracer ammunition should be stored separately from other ammunition.

c. If fired into or placed in a fire, small arms ammunition

does not explode violently. There are small individual explosions of each cartridge, the case flying in one direction and the bullet in another. In case of fire, it is advisable to keep those not engaged in fighting the fire at least 200 yards from the fire and have them lie on the ground. It is unlikely that the bullets and cases will fly over 200 yards. Small arms ammunition of all classes should be stored away from radiators, hot water pipes, and other sources of heat, since heat not only deteriorates ammunition, but also creates a fire hazard.

■ **72. BALLISTIC DATA.**—The approximate maximum ranges and average velocities for the several types and models of caliber .30 ammunition authorized for use in the machine gun, caliber .30, Browning, M1919A4, are shown below:

Type and model of caliber .30 cartridges	Average velocity 78 feet from muzzle (feet per second)	Approximate maximum range (yards)
Ball, M2	2,800	4,000
Ball, M1	2,600	5,500
Ball, M1906	2,640	3,450
Tracer, M1	2,700	3,500
Armor-piercing, M2	2,800	4,000
Blank, M1909		20
Dummy, M1906 (corrugated)		



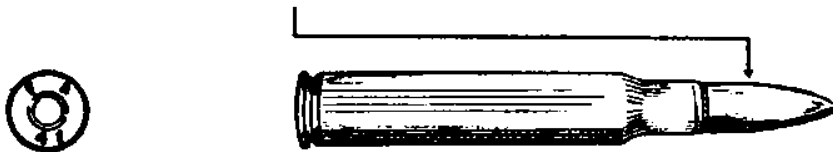
CAL. .30 ARMOR PIERCING AND TRACER CARTRIDGES

M1906-CUPRONICKEL

M1-GILDING METAL

M2-TIN COATED GILDING METAL (Prior to 1941)

M2-GILDING METAL (After 1940)



CAL. .30 BALL CARTRIDGES



CARTRIDGE, BLANK, CAL. .30, M1909



CARTRIDGE, DUMMY, CAL. .30 M1906 (Corrugated)

FIGURE 21.—Caliber .30 cartridges.

SECTION XI

CALIBER .22 TRAINING EQUIPMENT

■ 73. DESIGN AND DATA.—The caliber .22 machine-gun trainer M4 is provided for training purposes. It is used for firing the caliber .22 long rifle cartridge without modification or in any way affecting the standard gun. The caliber .22 trainer does not interfere with the mounting of the caliber .30 machine gun on the tripod M2.

a. Parts and accessories.—(1) The complete unit consists of a group of components and assemblies so designed that the caliber .30 gun may be converted to accommodate the caliber .22 long rifle cartridge merely by changing the main group assemblies (fig. 22). The following is a complete list of the component parts of the trainer and accessories for its use:

(a) *Parts.*

Bolt, assembly.
Firing pin, assembly.
Driving spring.
Cartridge stop.
Barrel.
Piston, actuating.
Cylinder, barrel.
Bearing, barrel, front.
Nut, barrel adjusting.
Nut, barrel locking.

(b) *Accessories.*

Belt, ammunition, including adapters and clips (par. 78).
Holders, cartridge.
Chute, cartridge holder.
Bag, metallic belt link.
Fixture, cartridge seating.
Tool, cartridge seating and ejecting.
Wrench, adjustable, 4-inch.

NOTE.—Safety of personnel is assured because the design is such that it is impossible to fire the caliber .30 cartridge with the caliber .22 trainer M4, assembly.

(2) Belt, adapters, and holders are loaded into a standard caliber .30 ammunition belt chest. A bag and chute are pro-